

COS30045

DATA VISUALISATION

TOPIC 02-1: DESIGN GUIDELINES TUFTE'S INTEGRITY PRINCIPLES PART 1

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Acknowledgment: This unit is partially based on the Visualisation unit developed by Alex Lex and Hanspeter Hfister (with permission).

Source of images used in this lecture available as links in the PDF version.



DESIGN GUIDELINES

What:

- ▶ design guidelines

Why:

- ▶ maintain integrity of the data, make sure users interpret/understand the data correctly

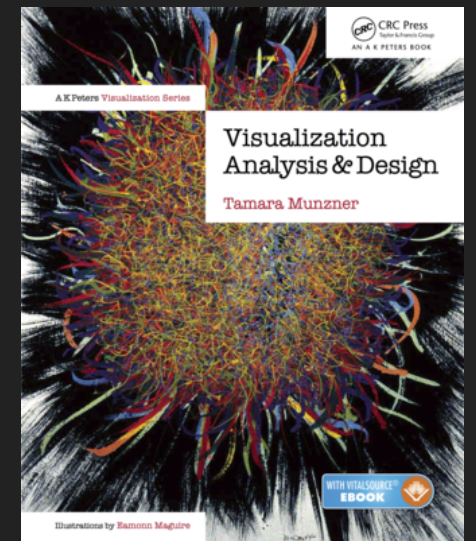
How:

- ▶ follow Tufte's integrity principles and visualisation design principles

DESIGN GUIDELINES

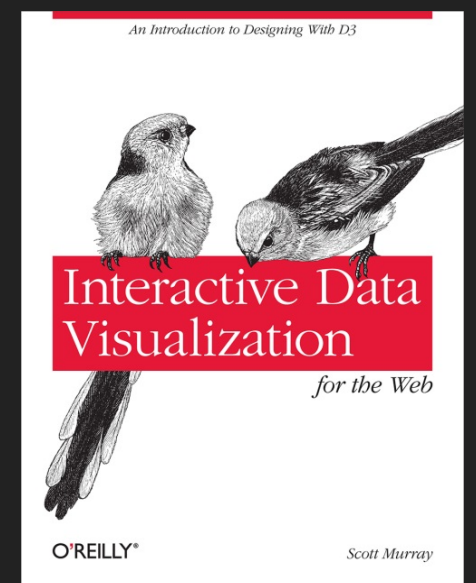
Lecture Readings

- ▶ VAD Ch 6
- ▶ Tufte (Book Ch on graphical integrity)



Tutorials

- ▶ IDV Ch 2 and 4



Well-designed presentations of interesting data are - a matter of *substance*, of *statistics*, and of *design*.

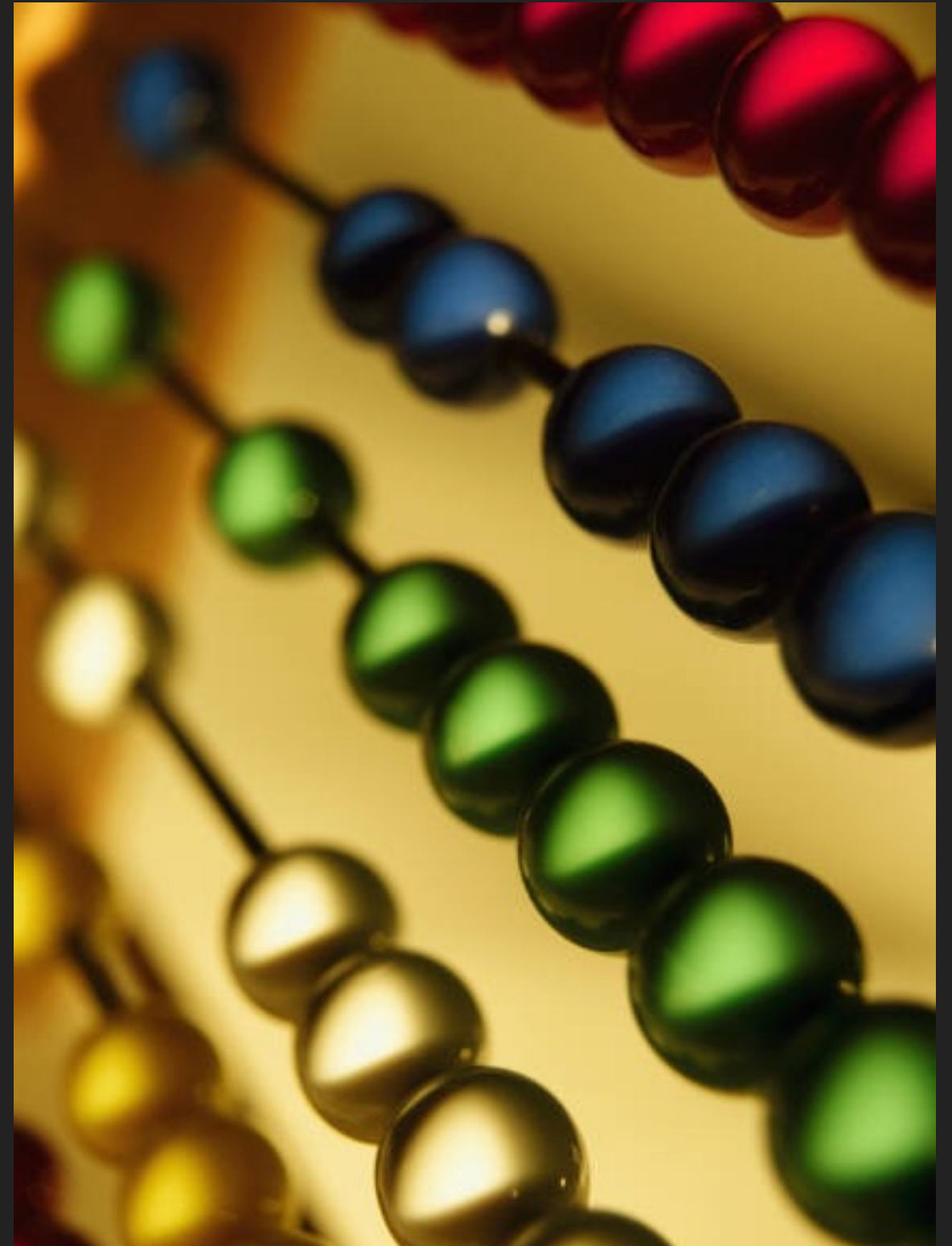
Edward Tufte

Statistician and data
visualisation guru

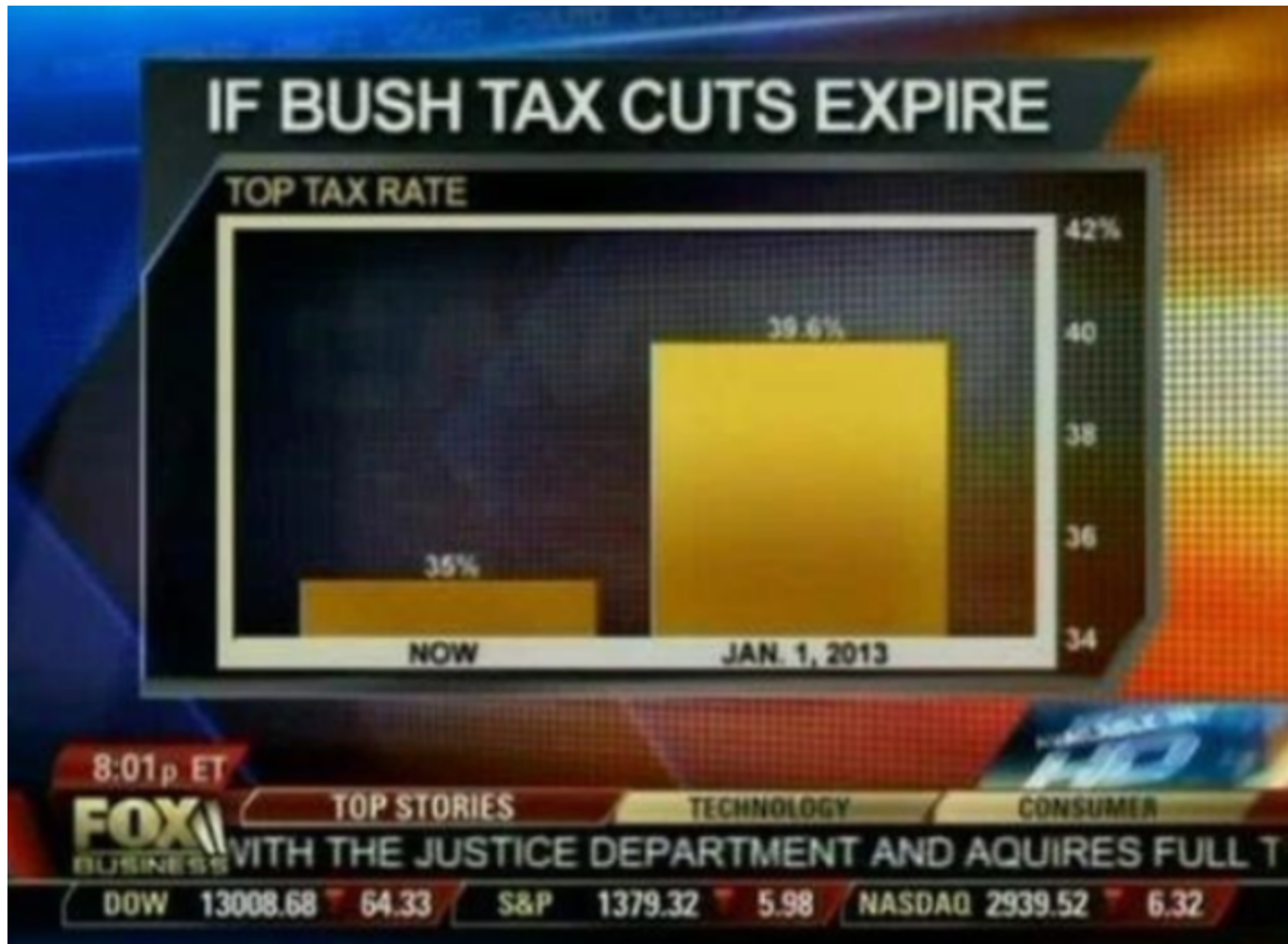


TUFTE'S INTEGRITY PRINCIPLES

- ▶ Clear, detailed, and thorough labelling to prevent graphical distortion and ambiguity
- ▶ Size of the physically measured graphic should be directly proportional to the numerical quantities ("lie factor")
- ▶ Show data variation, not design variation
- ▶ Graphics must not quote data out of context

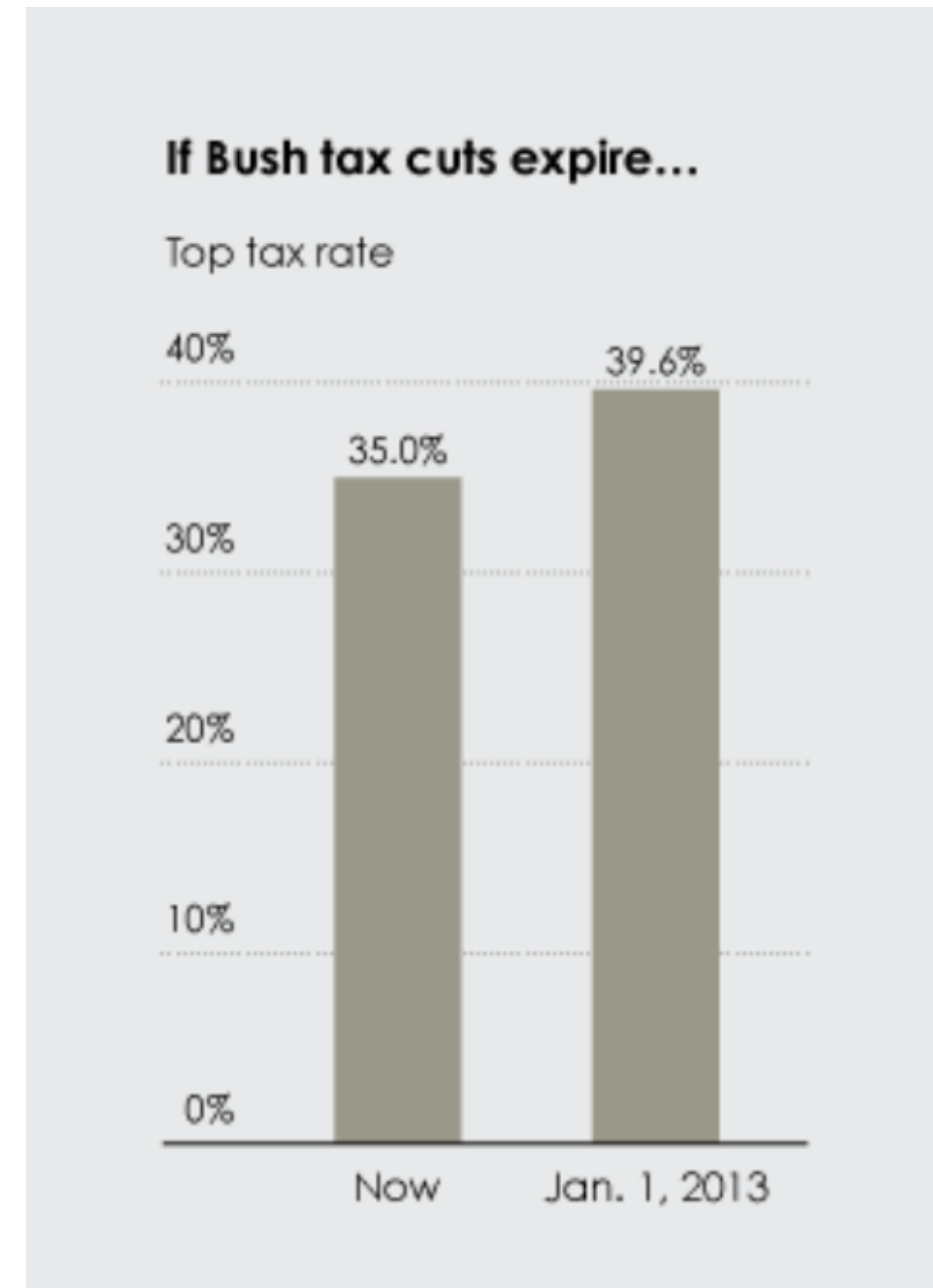


INTEGRITY PRINCIPLE: PREVENT GRAPHICAL DISTORTION

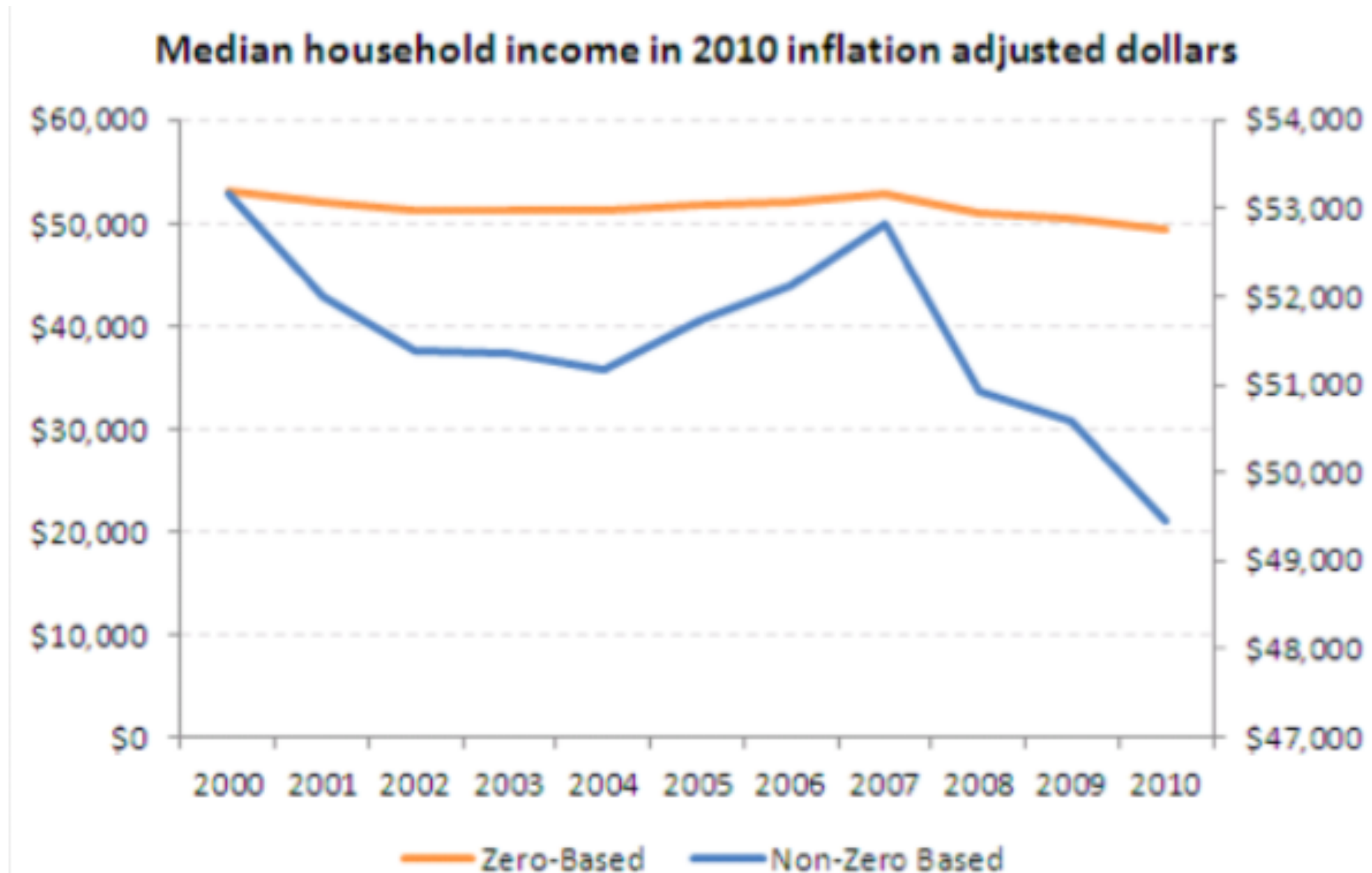


INTEGRITY PRINCIPLE: PREVENT GRAPHICAL DISTORTION

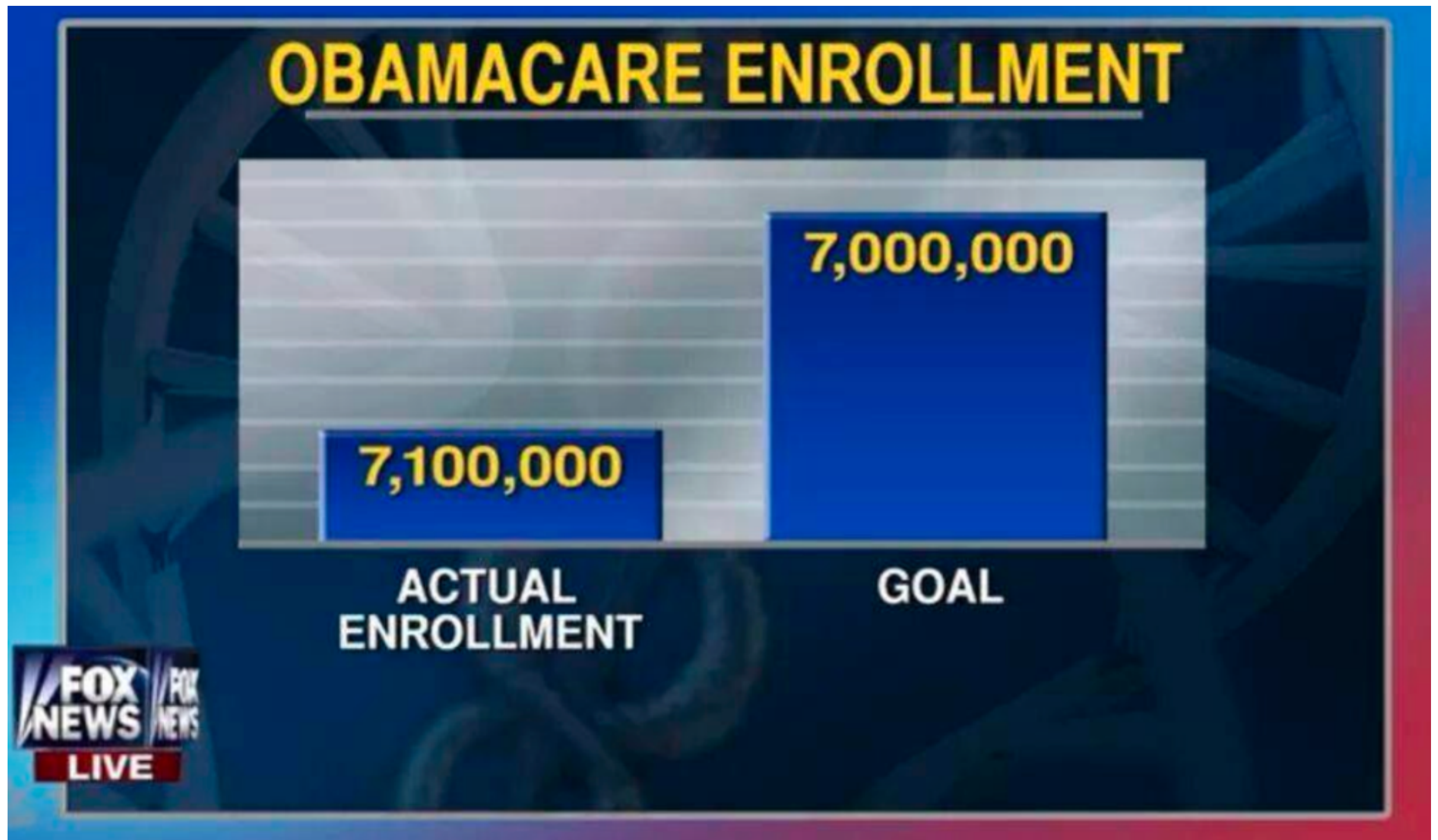
- ▶ How do we judge the size of effect in bar graph?
 - ▶ difference in data must match difference in graph (see lie factor)
 - ▶ relative size of bars is important
 - ▶ **ALWAYS START BAR CHARTS AT 0**



INTEGRITY PRINCIPLE: NON-ZERO SCALE DISTORTIONS



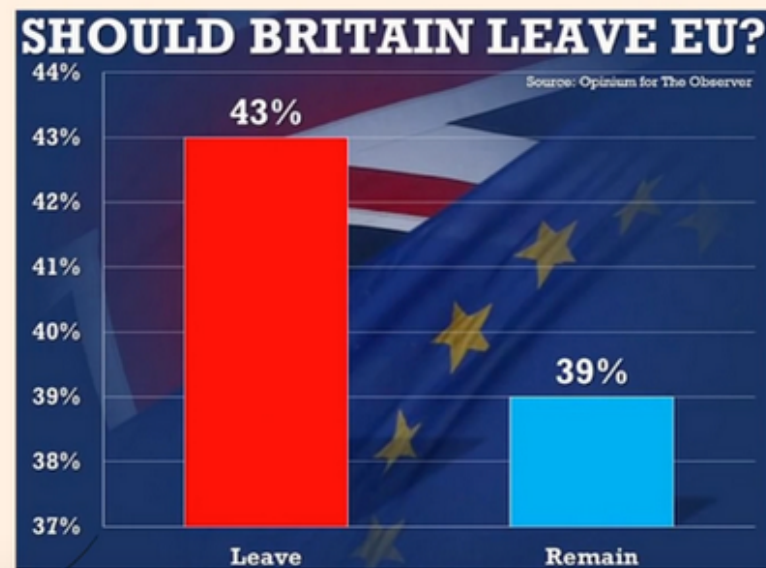
INTEGRITY PRINCIPLE: PREVENT GRAPHICAL DISTORTION



INTEGRITY PRINCIPLE: PREVENT GRAPHICAL DISTORTION

Graphics that are accurate but misleading

Baseline should start at zero, not 37



Graphics that are accurate but misleading

A better chart of the same data

Should Britain leave the EU?

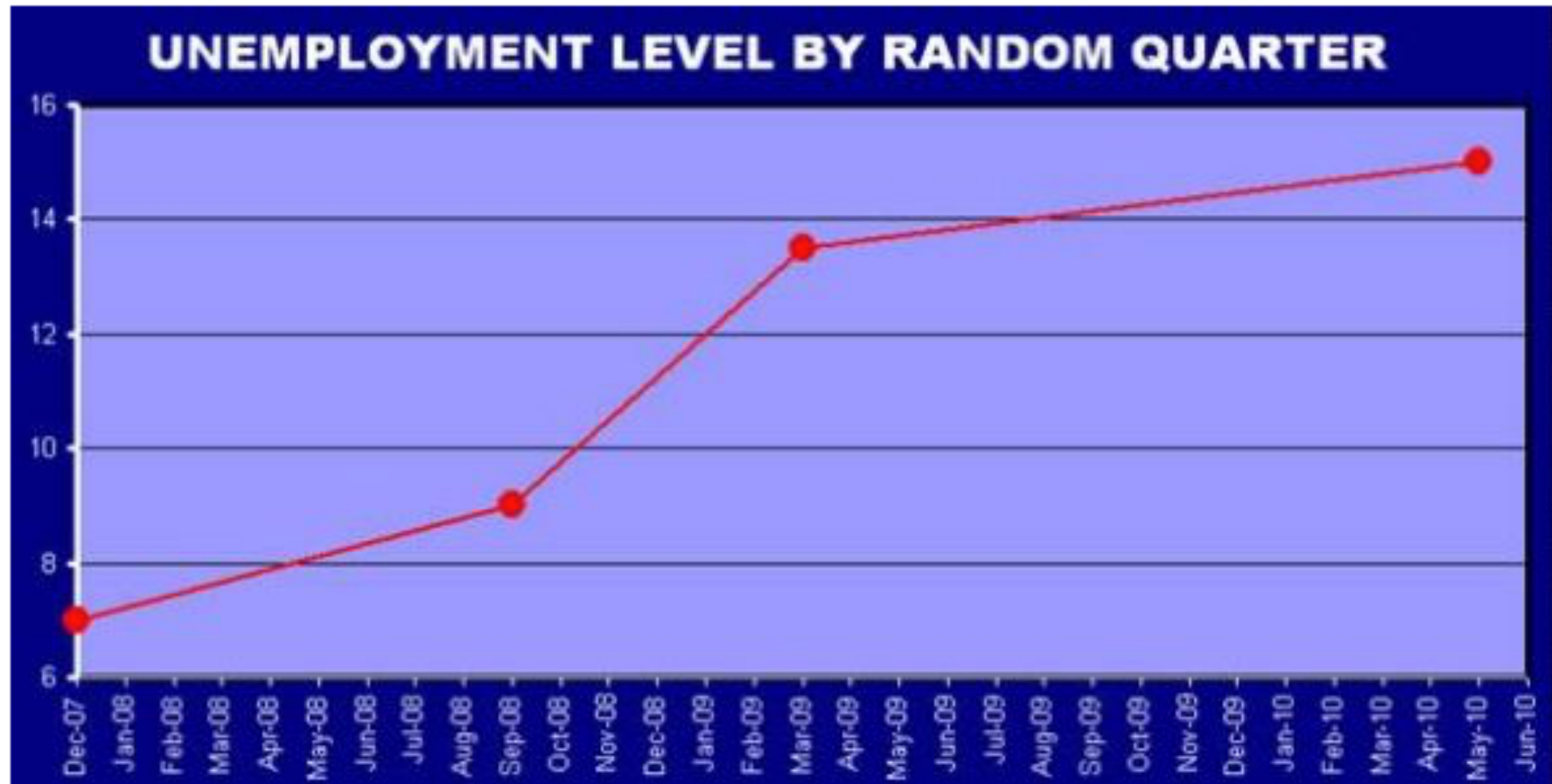


INTEGRITY PRINCIPLE: PREVENT GRAPHICAL DISTORTION



Manipulation of x-axis (Sep to Mar = 6 mths, Mar to Jun 15 mths)

INTEGRITY PRINCIPLE: PREVENT GRAPHICAL DISTORTION

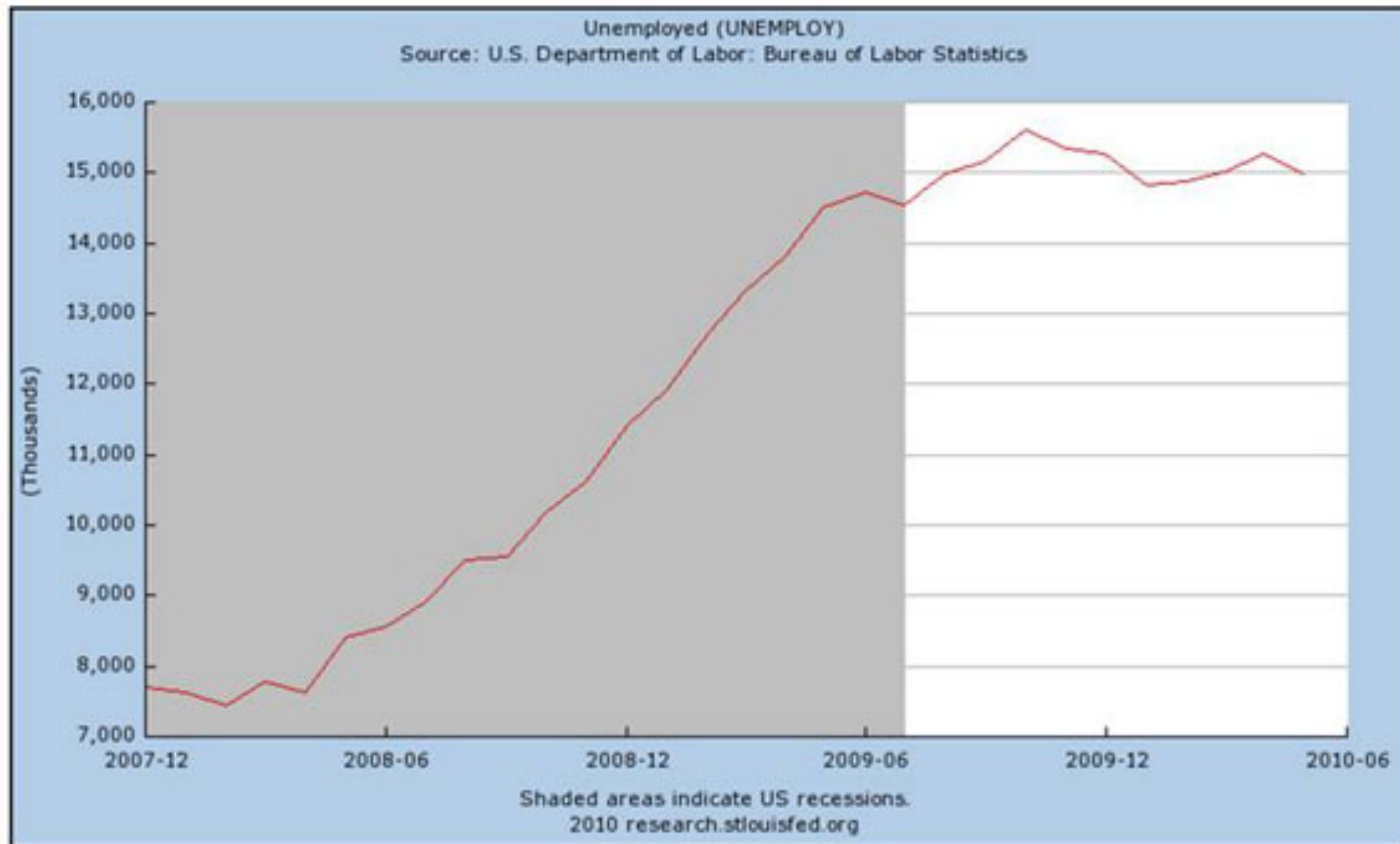


Data is for unemployment, not job losses

Data is for 'random' quarters

Manipulation of x-axis (Sep to Mar = 6 mths, Mar to Jun 15 mths)

INTEGRITY PRINCIPLE: SCALE DISTORTIONS – OUT OF CONTEXT



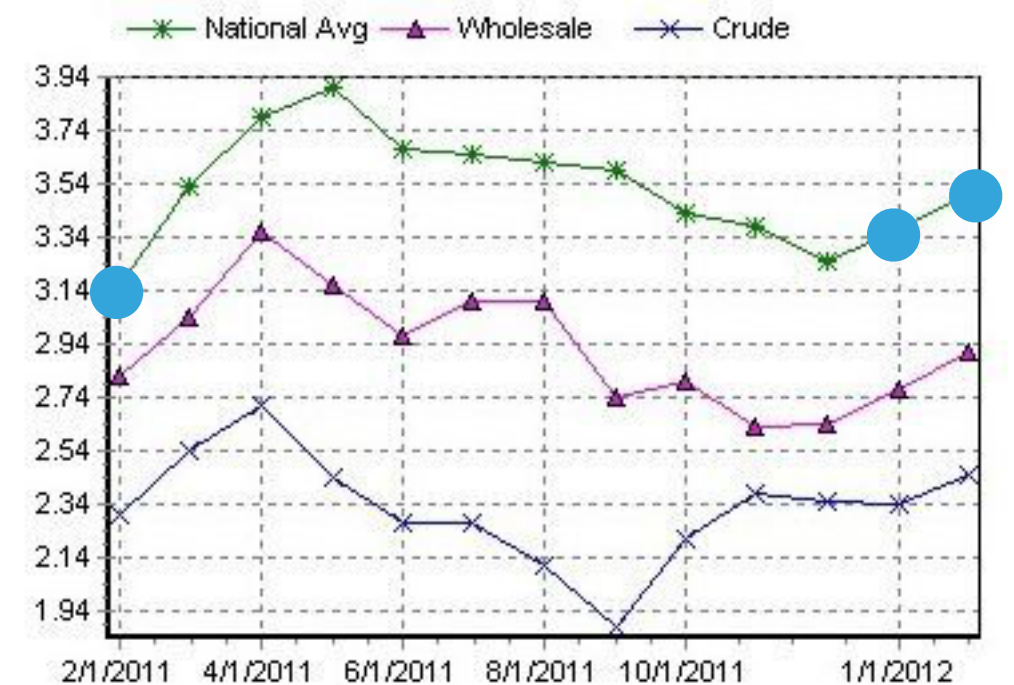
INTEGRITY PRINCIPLE: SCALE DISTORTIONS – OUT OF CONTEXT



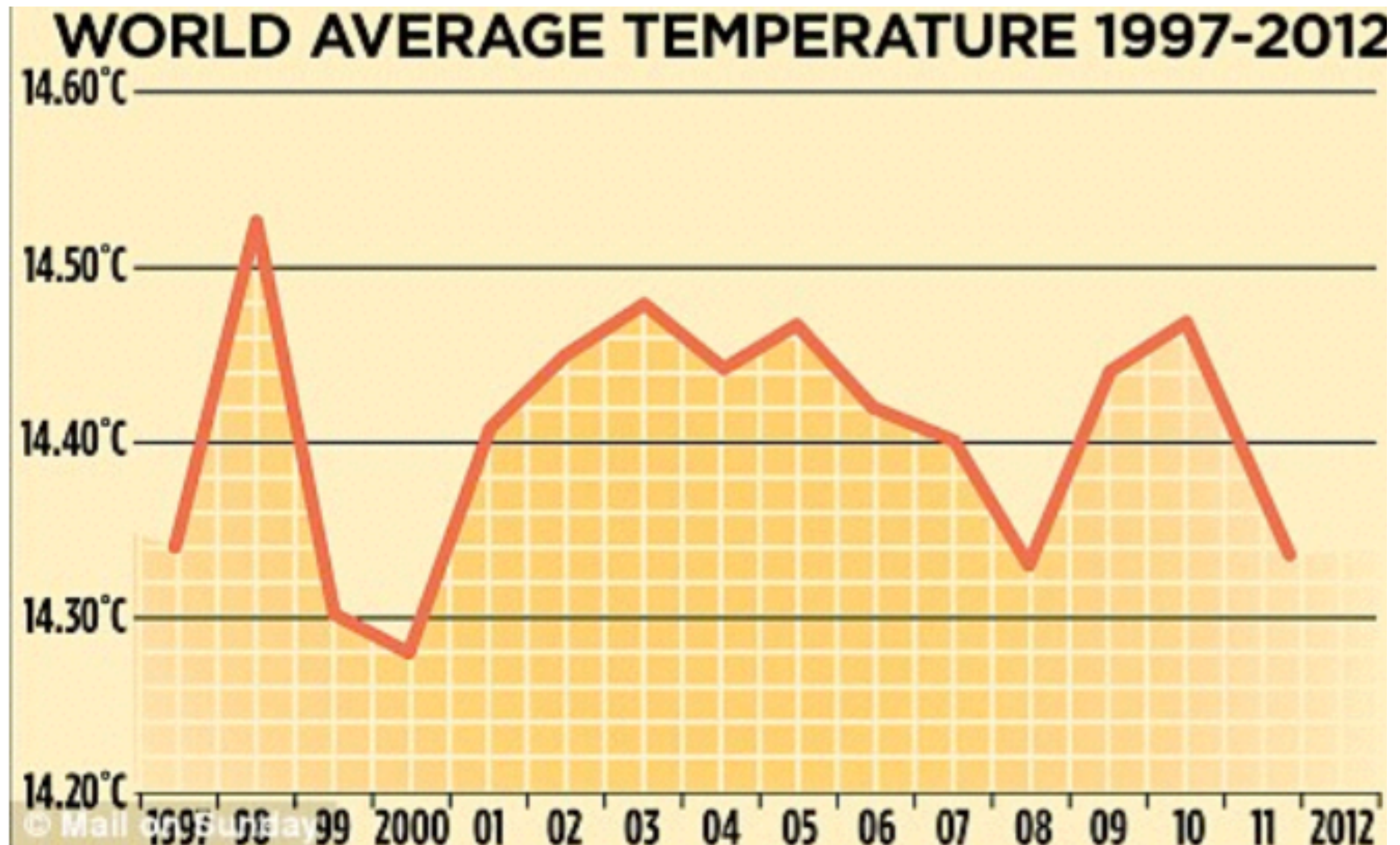
INTEGRITY PRINCIPLE: SCALE DISTORTIONS – OUT OF CONTEXT



12 Month Average for Self-Serve Regular



INTEGRITY PRINCIPLE: SCALE DISTORTIONS – OUT OF CONTEXT

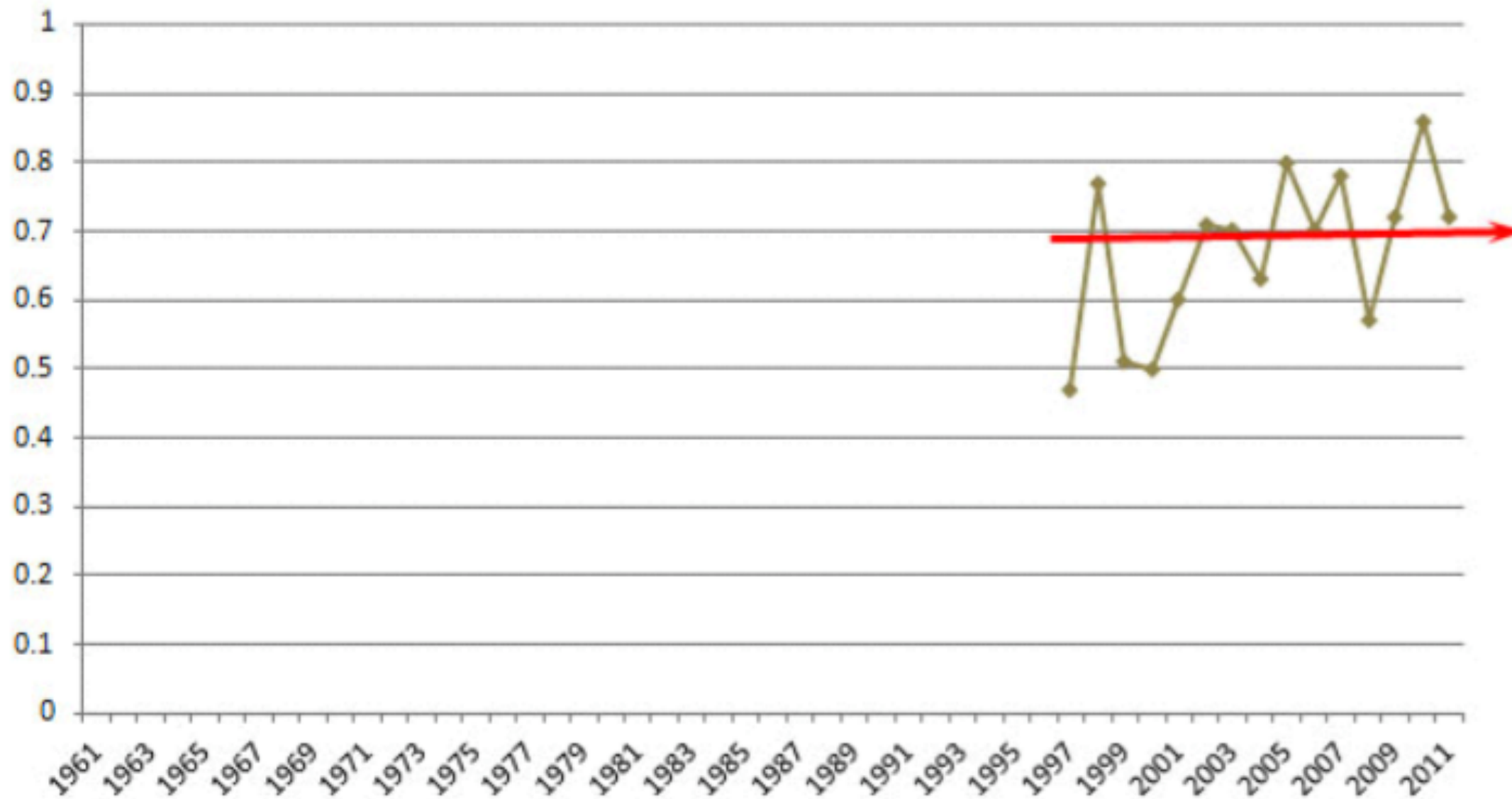


Does not appear to be a warming trend...

Daily Mail

INTEGRITY PRINCIPLE: SCALE DISTORTIONS

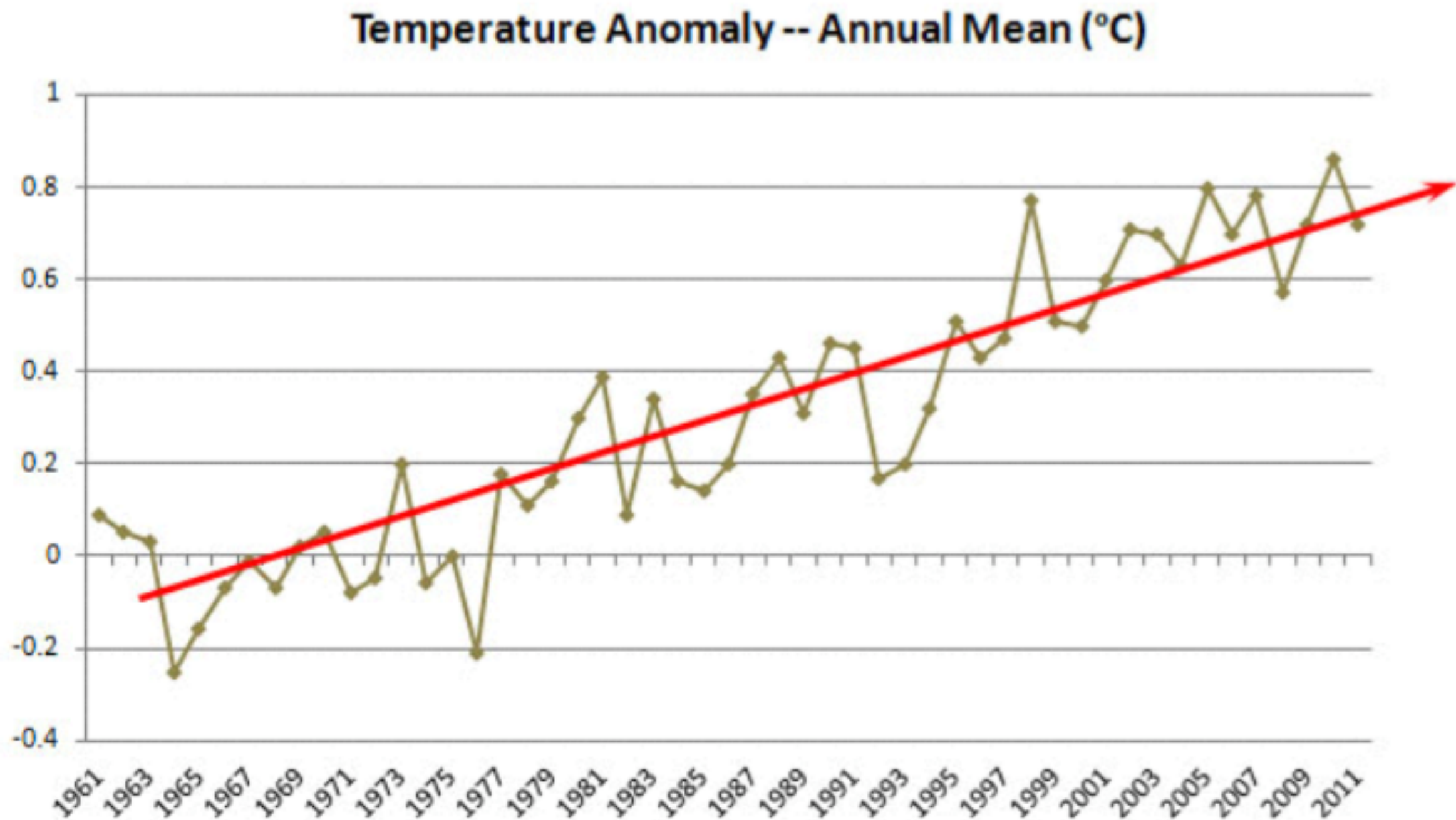
Temperature Anomaly -- Annual Mean (°C)



Does not appear to be a warming trend, until...

15 year data plotted by Daily Mail
(source Mother Jones)

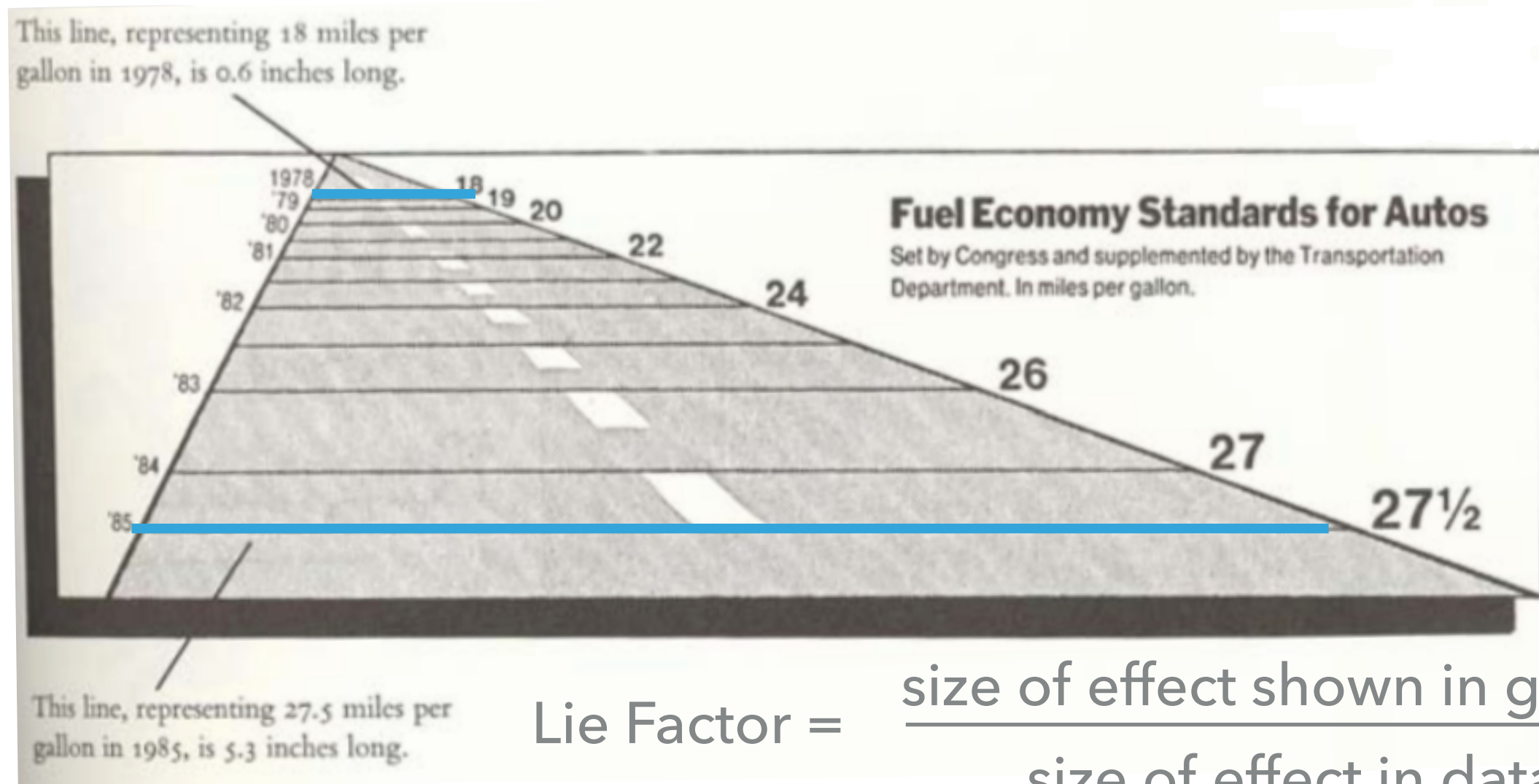
INTEGRITY PRINCIPLE: SCALE DISTORTIONS



...you consider a larger data set

50 year data set
(source Mother Jones)

INTEGRITY PRINCIPLE: SCALE DISTORTIONS – THE LIE FACTOR



1.5 cm = 29 km

13.5 cm = 44 km

2.9 cm = 29 km

4.4 cm = 44 km

INTEGRITY PRINCIPLE: SCALE DISTORTIONS



Austrian welfare payments

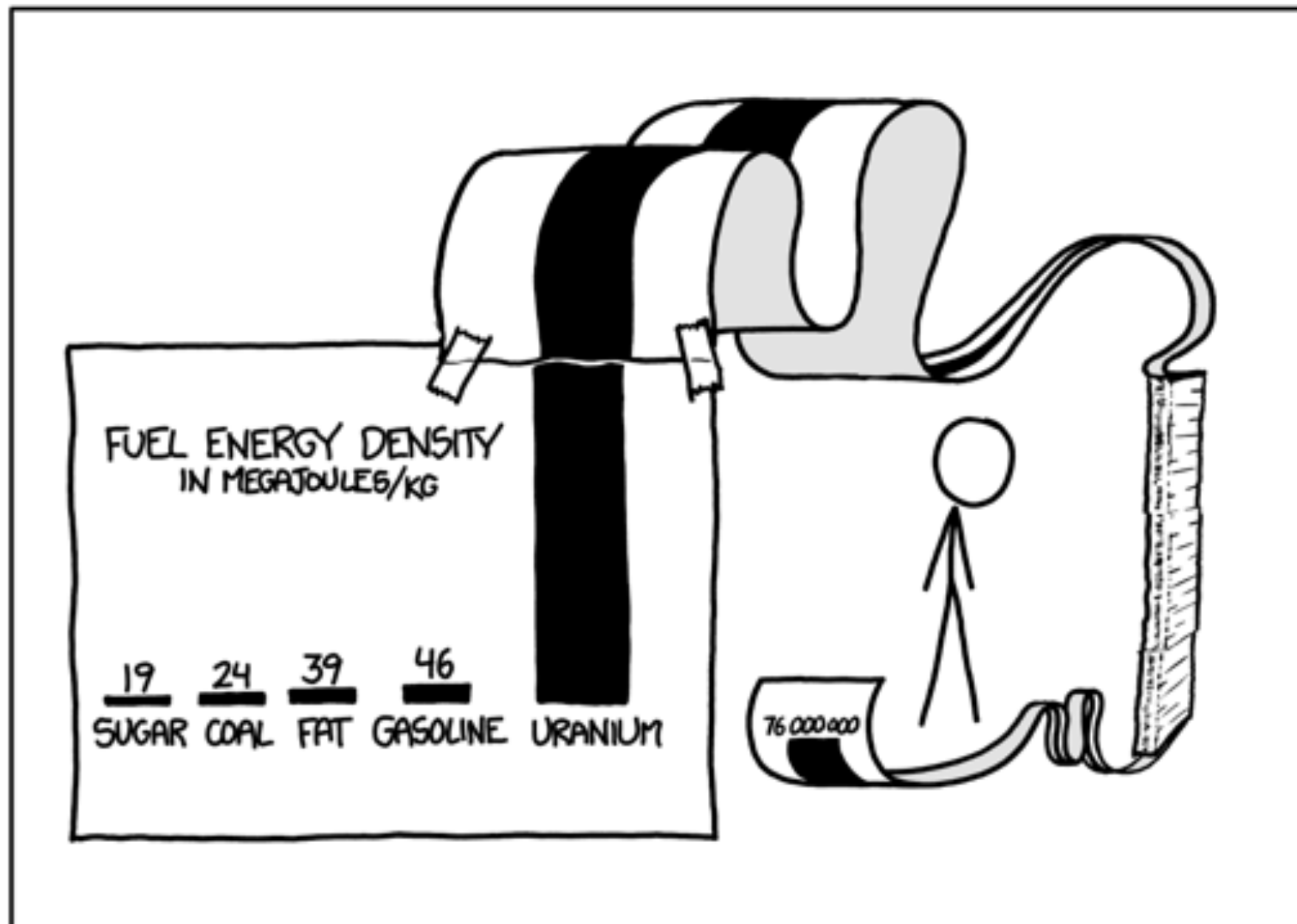
INTEGRITY PRINCIPLE: SCALE DISTORTIONS

**Grafik der
Kronen-
zeitung**



**Grafik
in echt**





SCIENCE TIP: LOG SCALES ARE FOR QUITTERS WHO CAN'T FIND ENOUGH PAPER TO MAKE THEIR POINT *PROPERLY*.

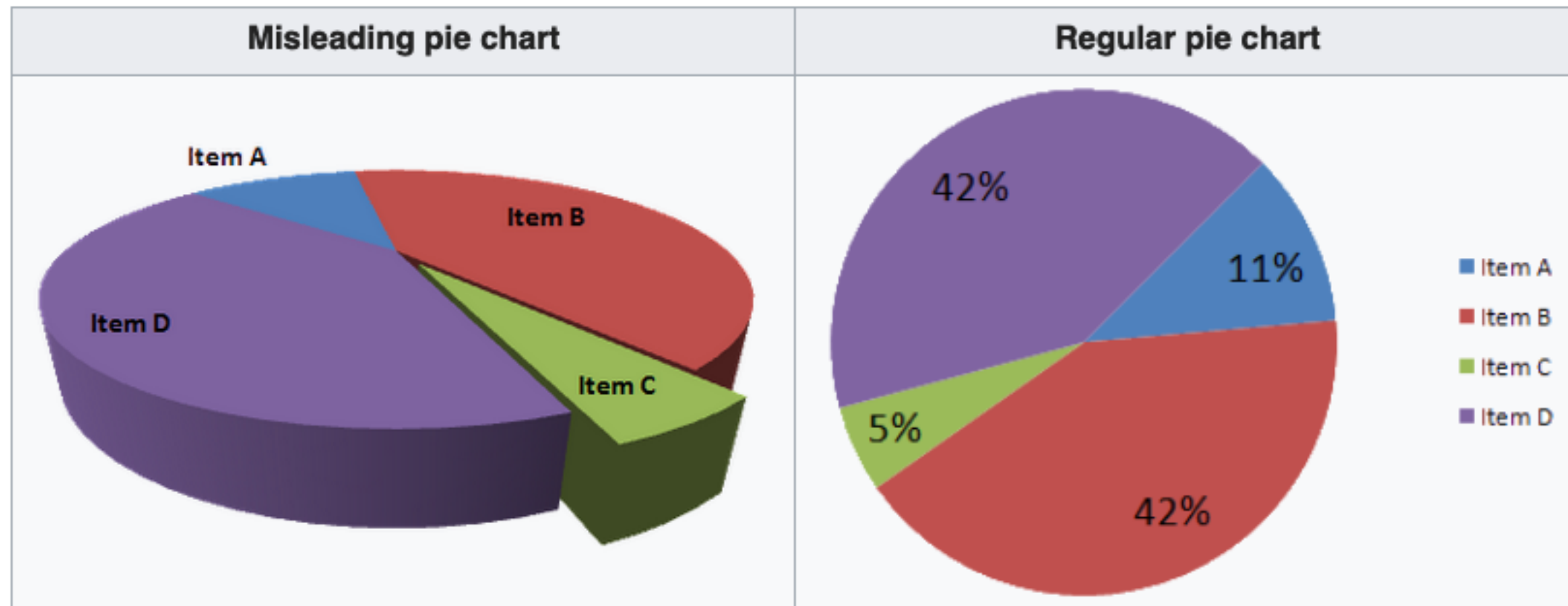
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DATA VISUALISATION

TOPIC 02-2: DESIGN GUIDELINES TUFTE'S INTEGRITY PRINCIPLES PART 2

INTEGRITY PRINCIPLE: SCALE DISTORTIONS IN 3D

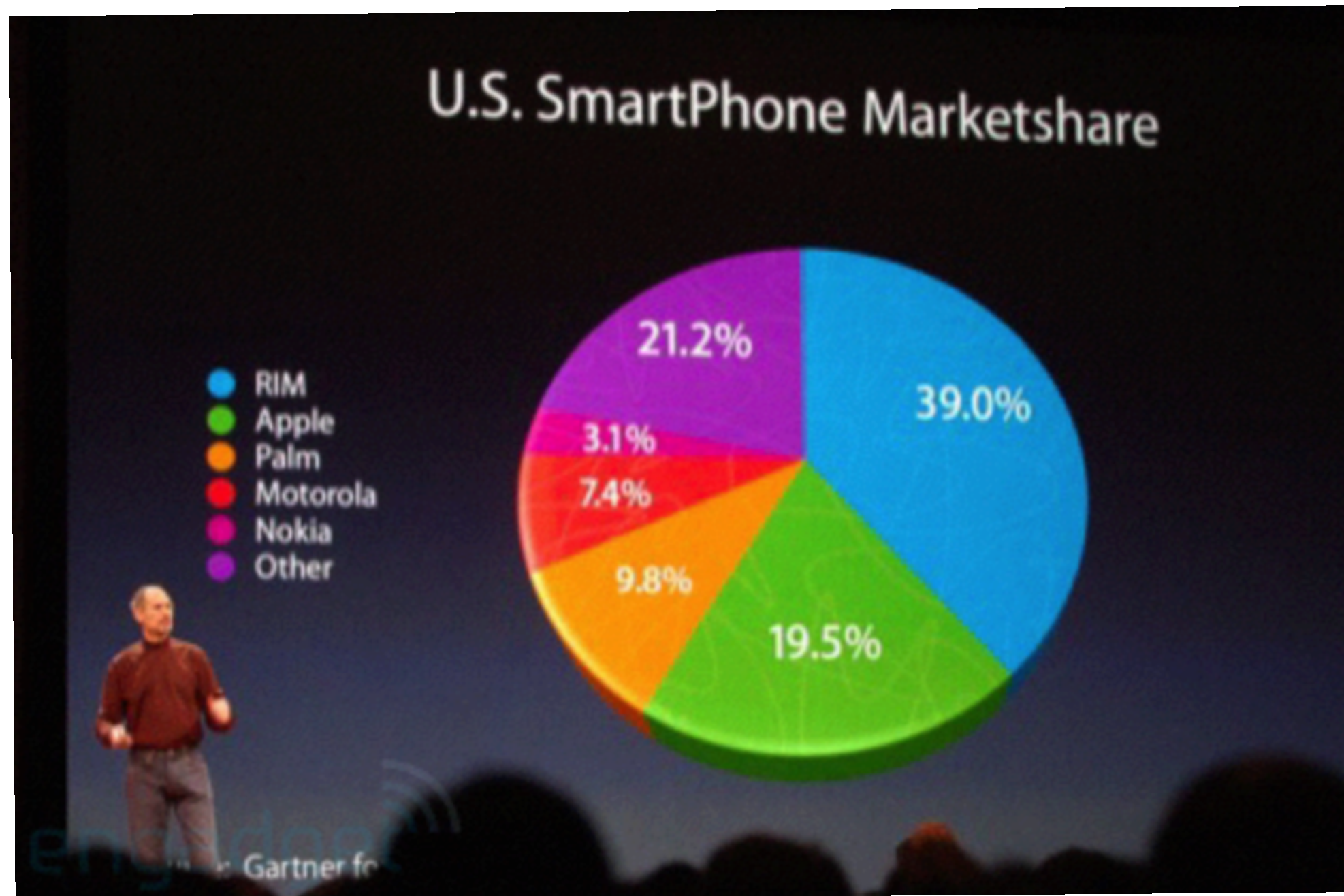
Comparison of pie charts



3D Pie Chart

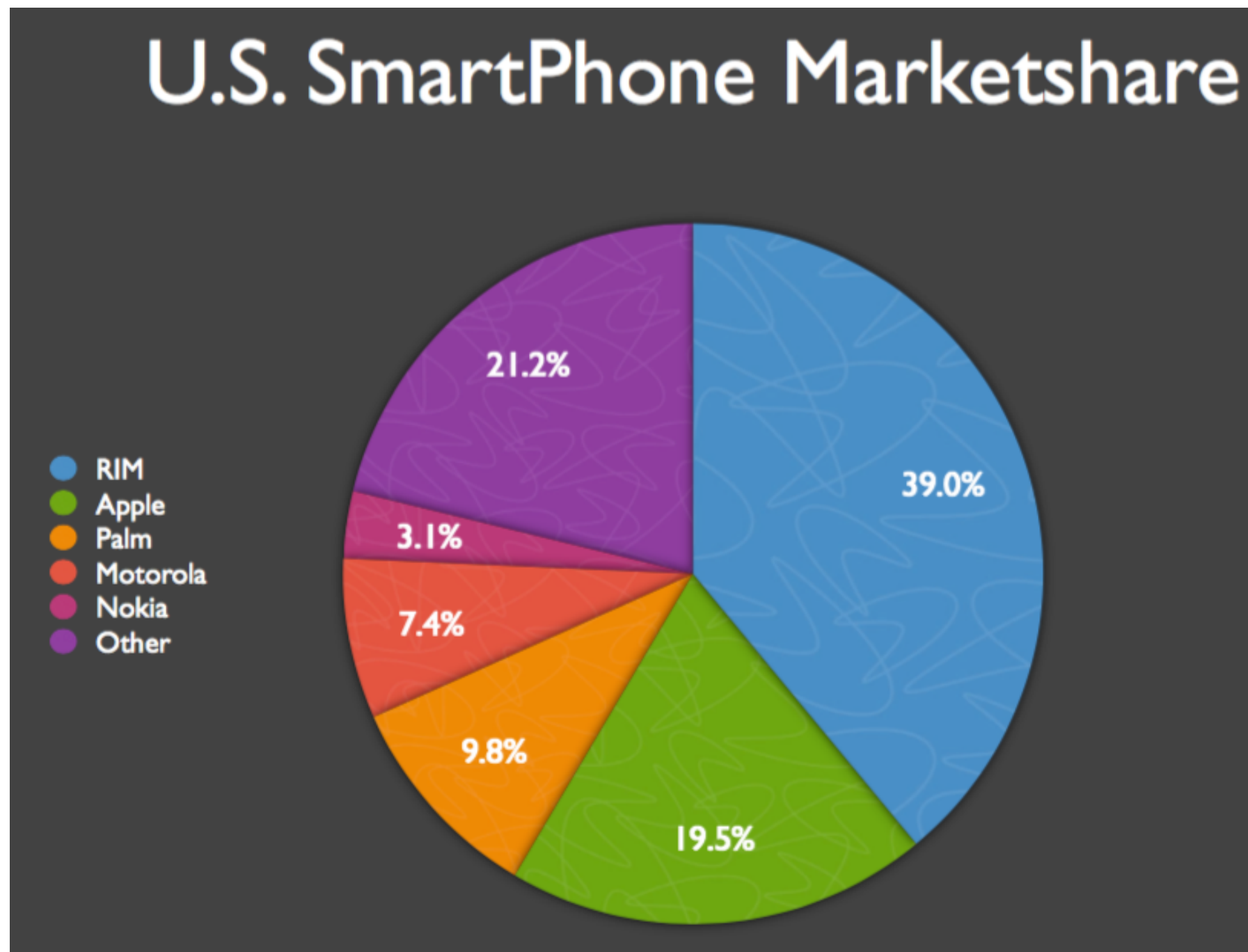
- ▶ perspective - things at front look bigger than those at the back
- ▶ difficult to judge the relative magnitude

INTEGRITY PRINCIPLE: SCALE DISTORTIONS IN 3D



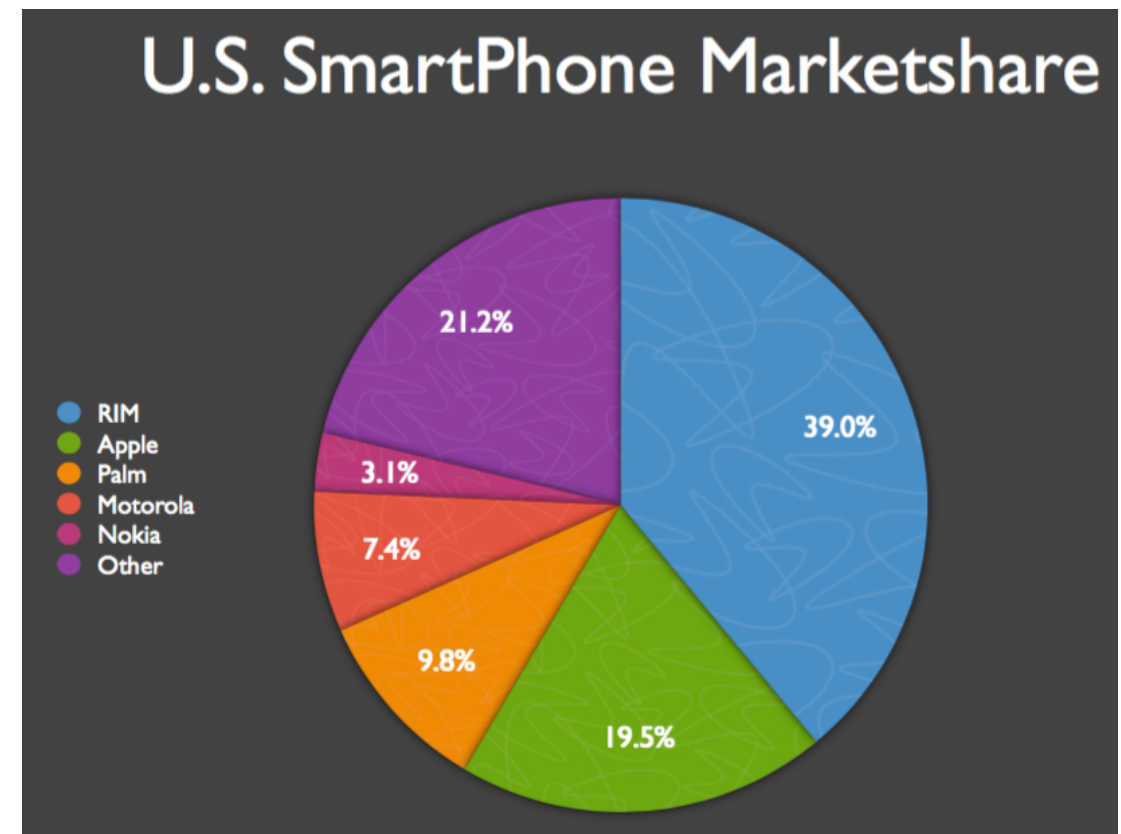
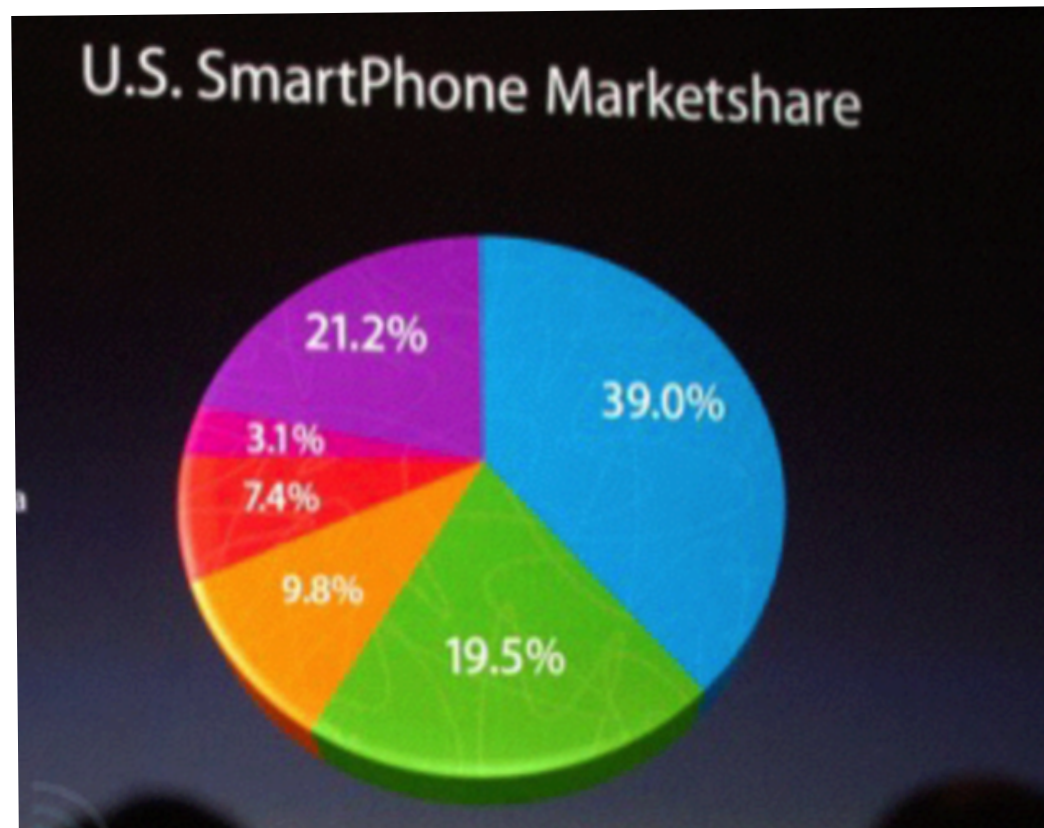
How far behind RIM is Apple?

INTEGRITY PRINCIPLE: SCALE DISTORTIONS IN 3D



How far behind RIM is Apple?

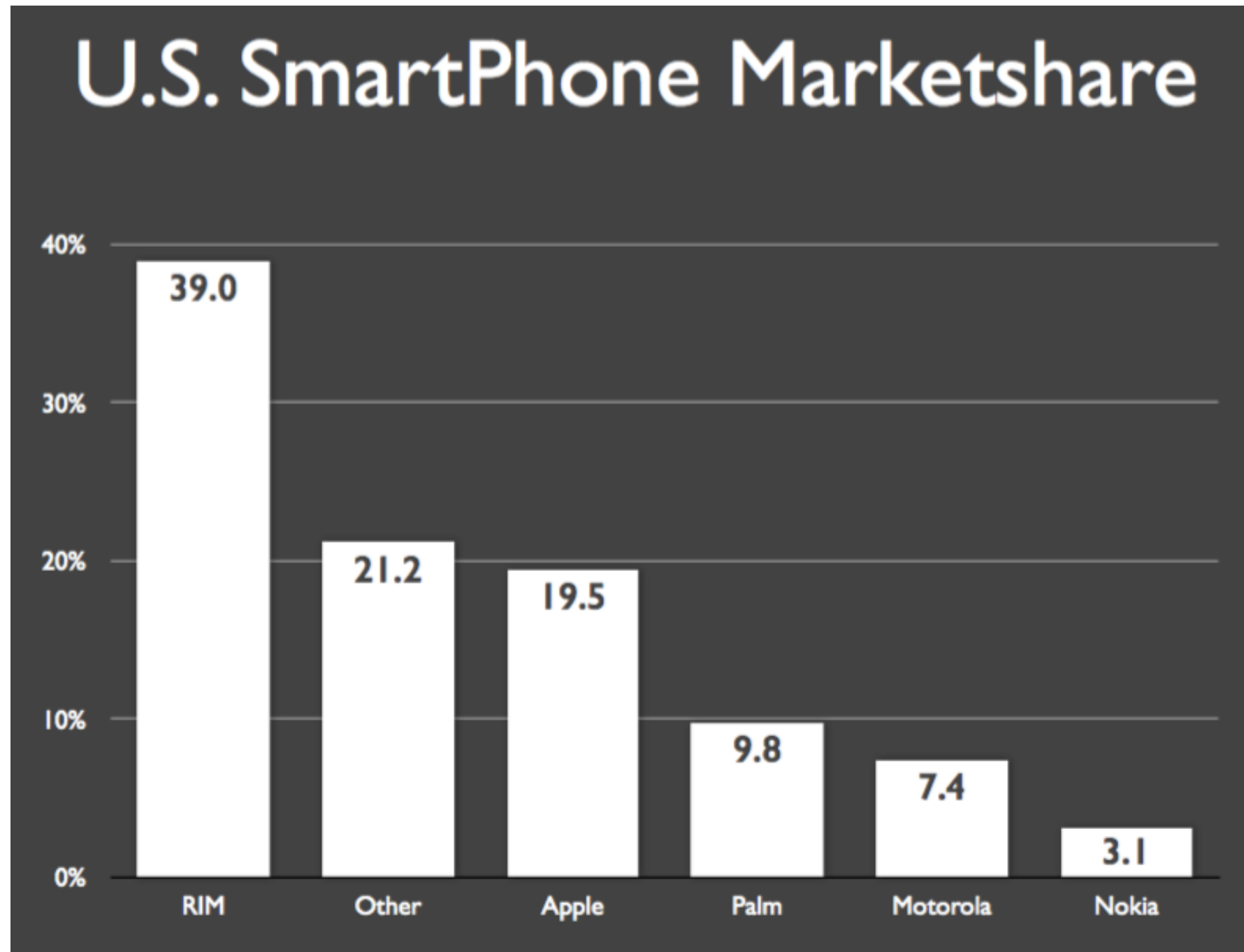
INTEGRITY PRINCIPLE: SCALE DISTORTIONS IN 3D



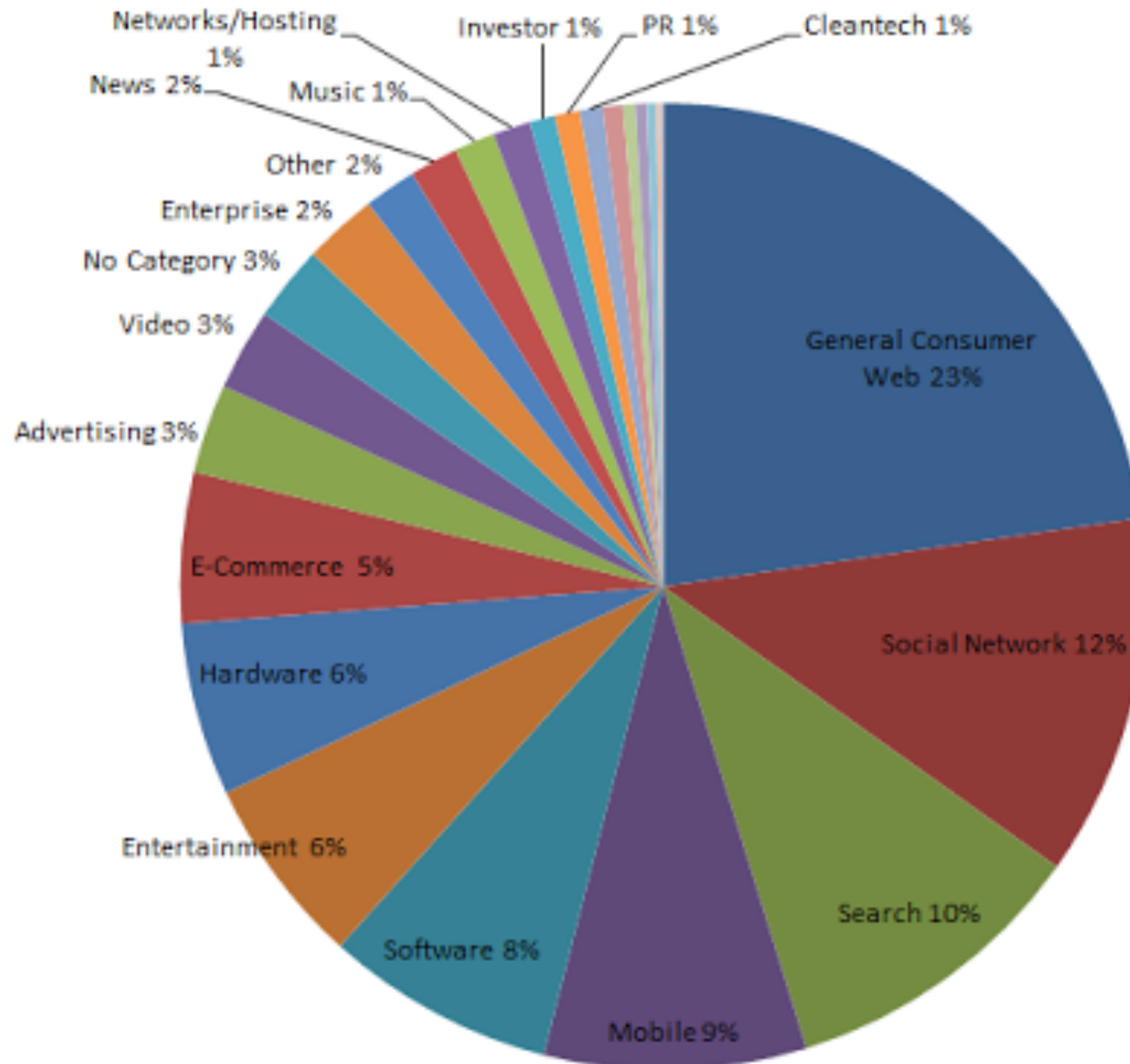
Slice at the front of the pie appears larger than it is

Slice in the back appears smaller than it is

BAR GRAPH

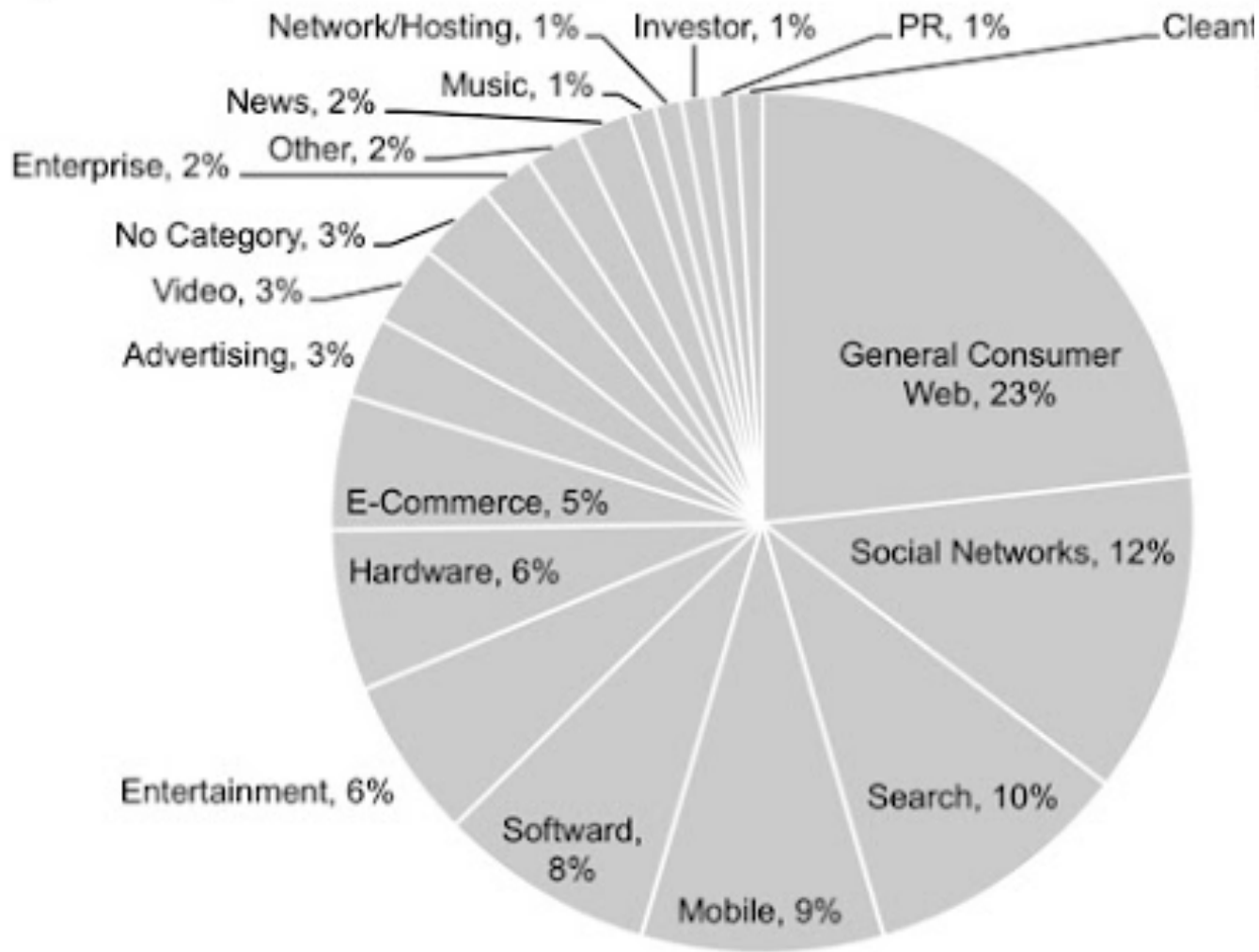


PIE CHART PROBLEMS



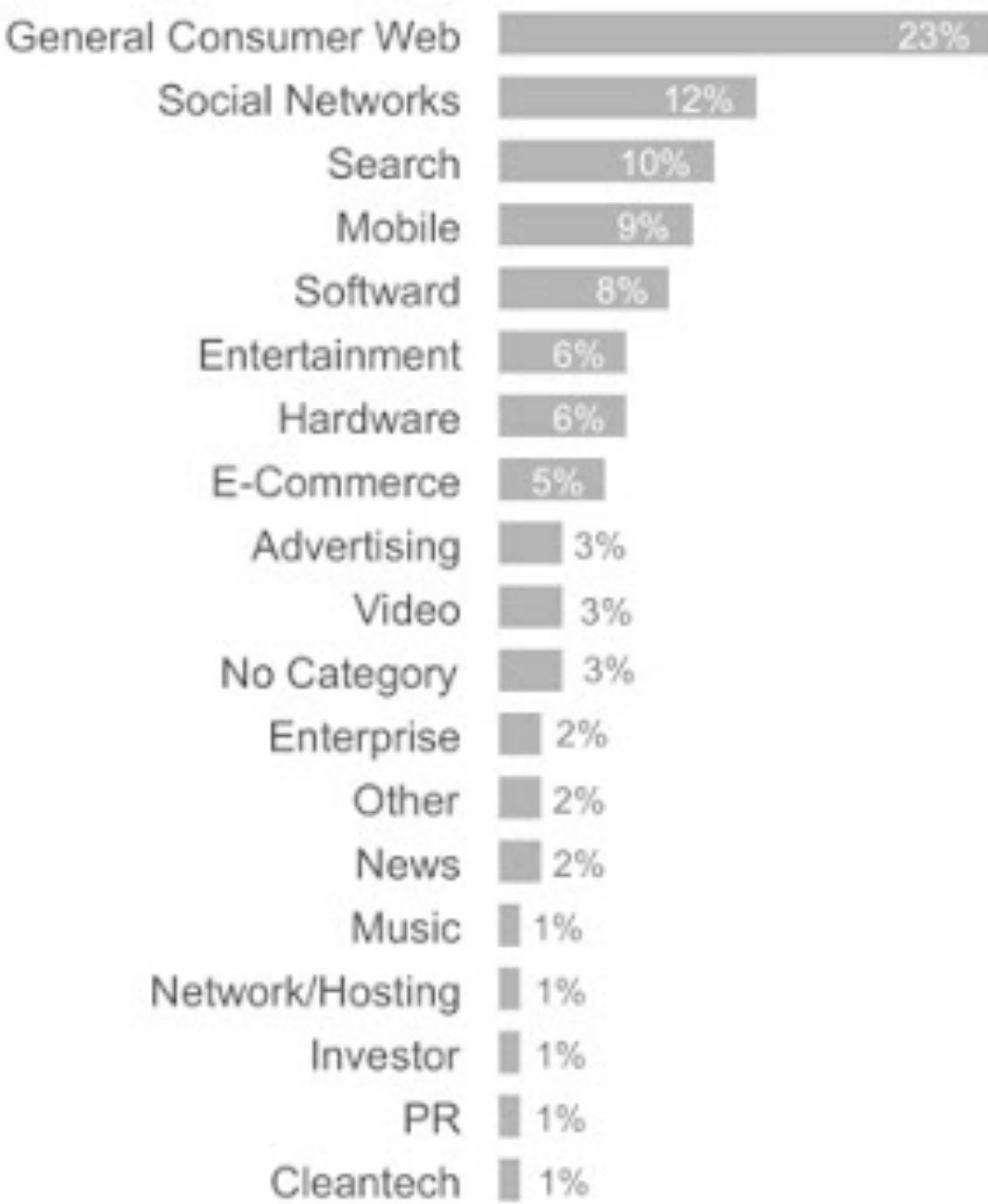
TechCrunch Coverage: 2005 - 2011

A slightly better pie?

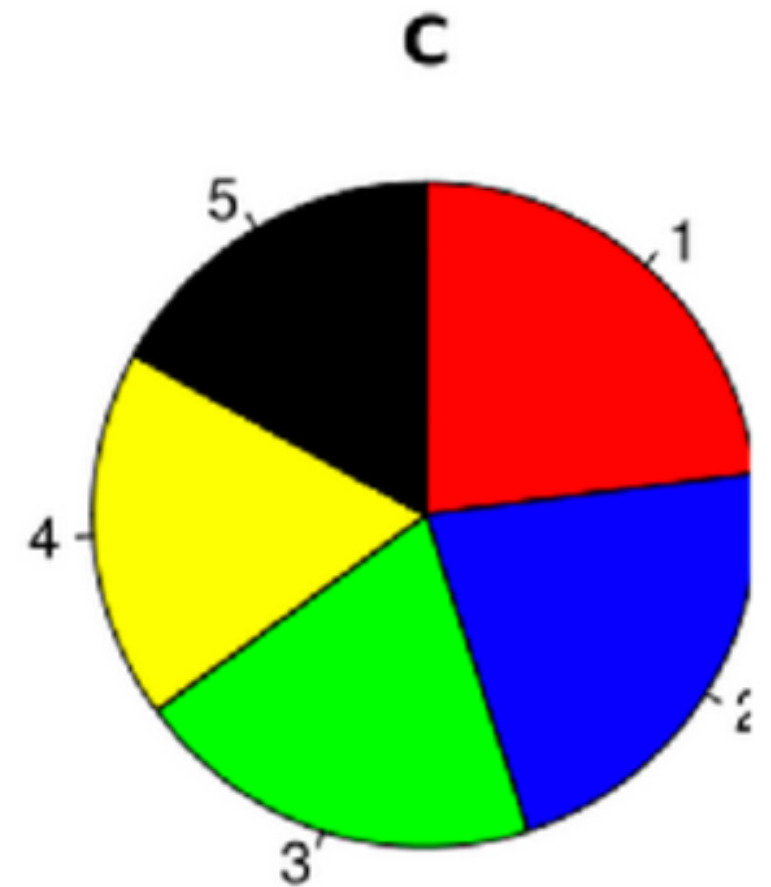


TechCrunch Coverage: 2005 - 2011

Bars are best!

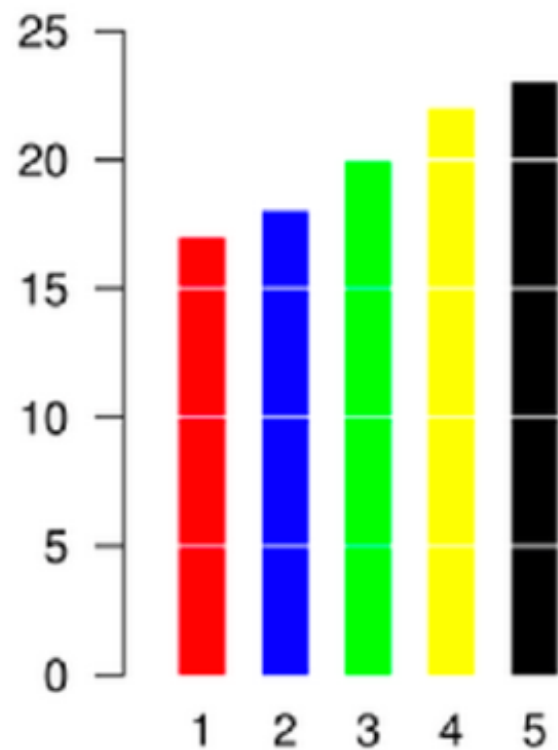


PIE CHART INTERPRETATION – BIGGEST TO SMALLEST?

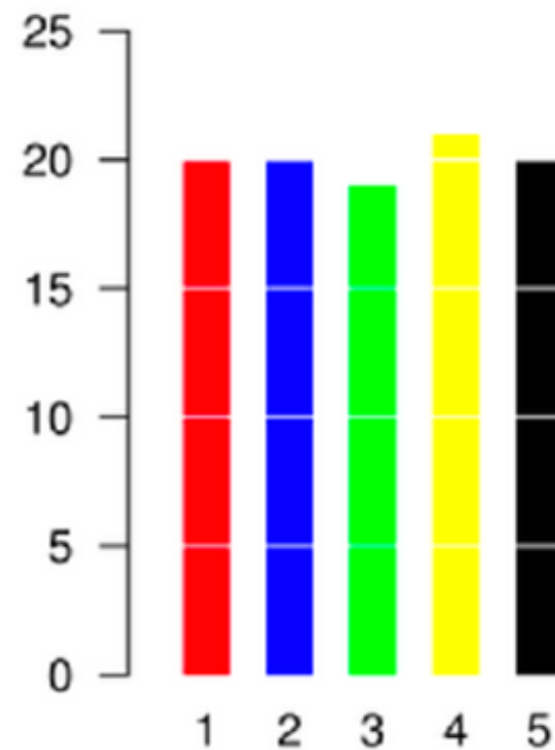
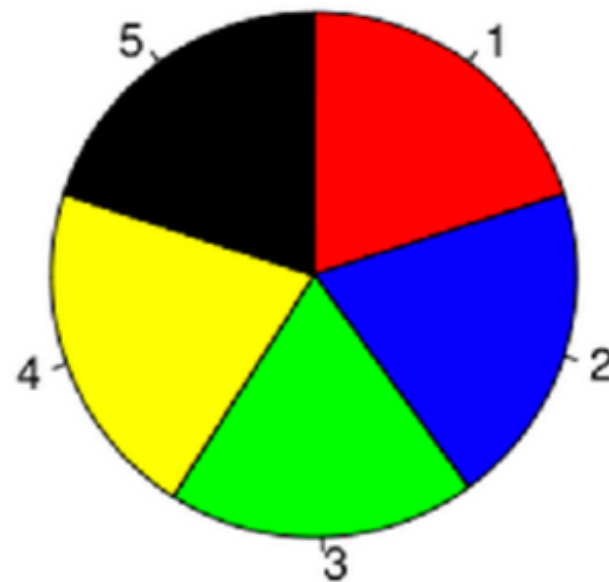


PIE CHART INTERPRETATION – BIGGEST TO SMALLEST?

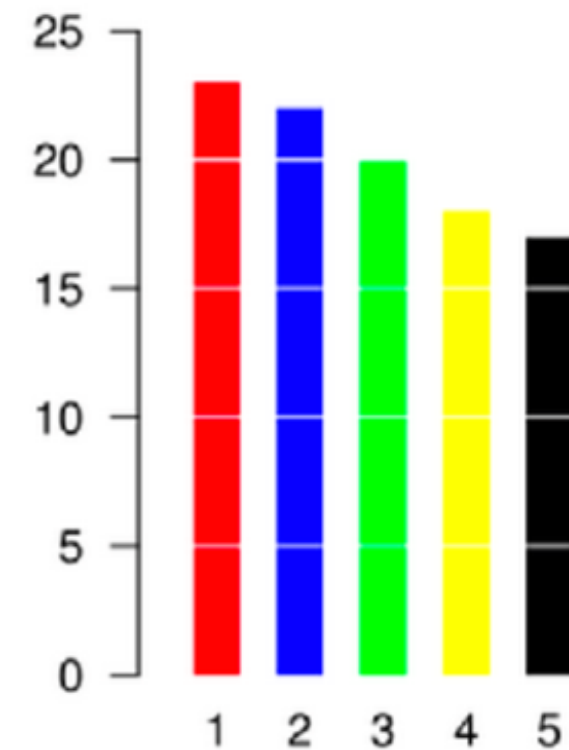
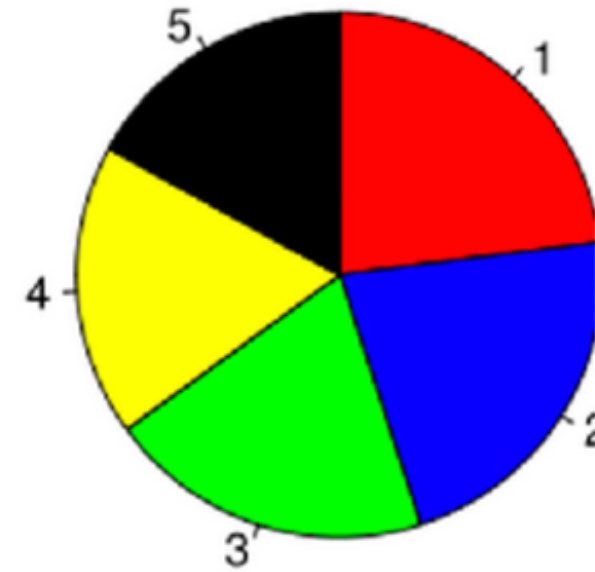
A



B



C

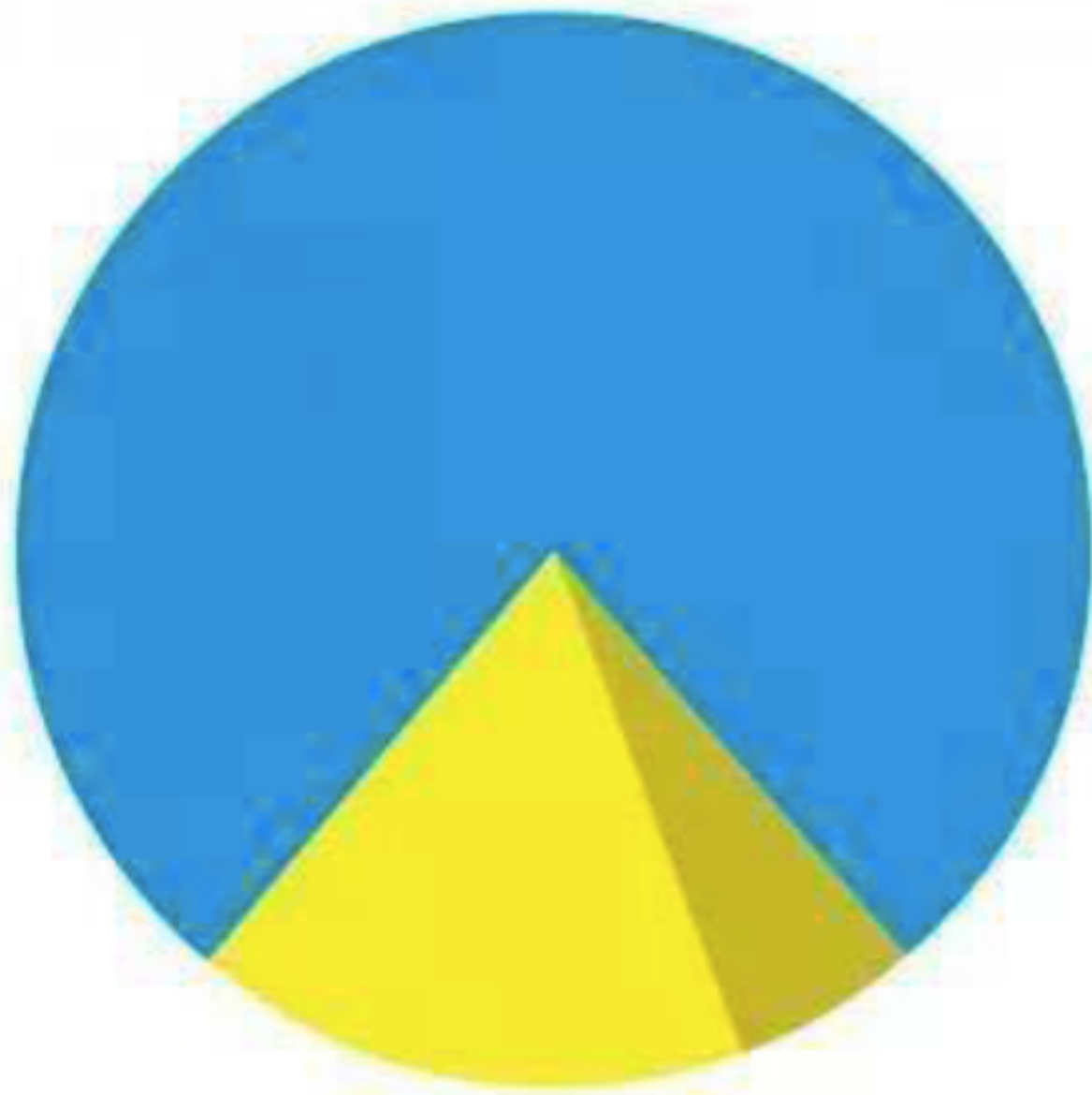


PIE PROBLEMS

Convictions in England and Wales for class A drug supply.



THE DATA VIS FAVOURITE PIE CHART



Sky



Sunny side of pyramid



Shady side of pyramid

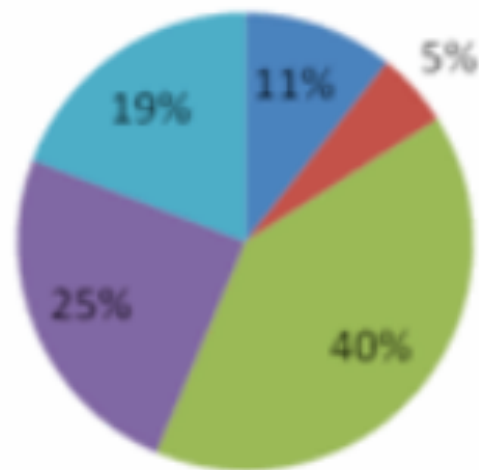
THE DATA VIS SECOND FAVOURITE PIE CHART



COMMUNICATION – WHAT IS THE IMPORTANT BIT OF INFORMATION?

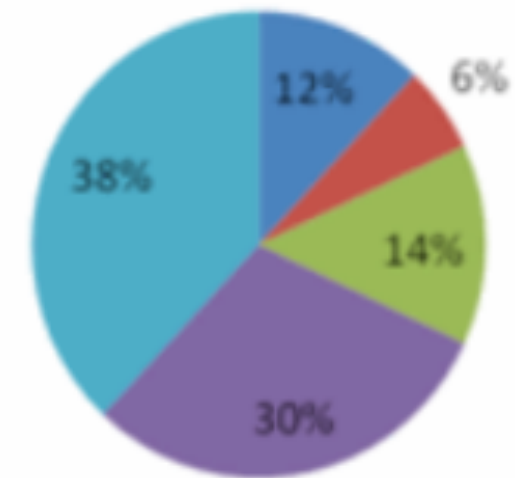
PRE: How do you feel about doing science?

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited



POST: How do you feel about doing science?

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited



REPORT THE IMPORTANT NUMBER

After the pilot program,

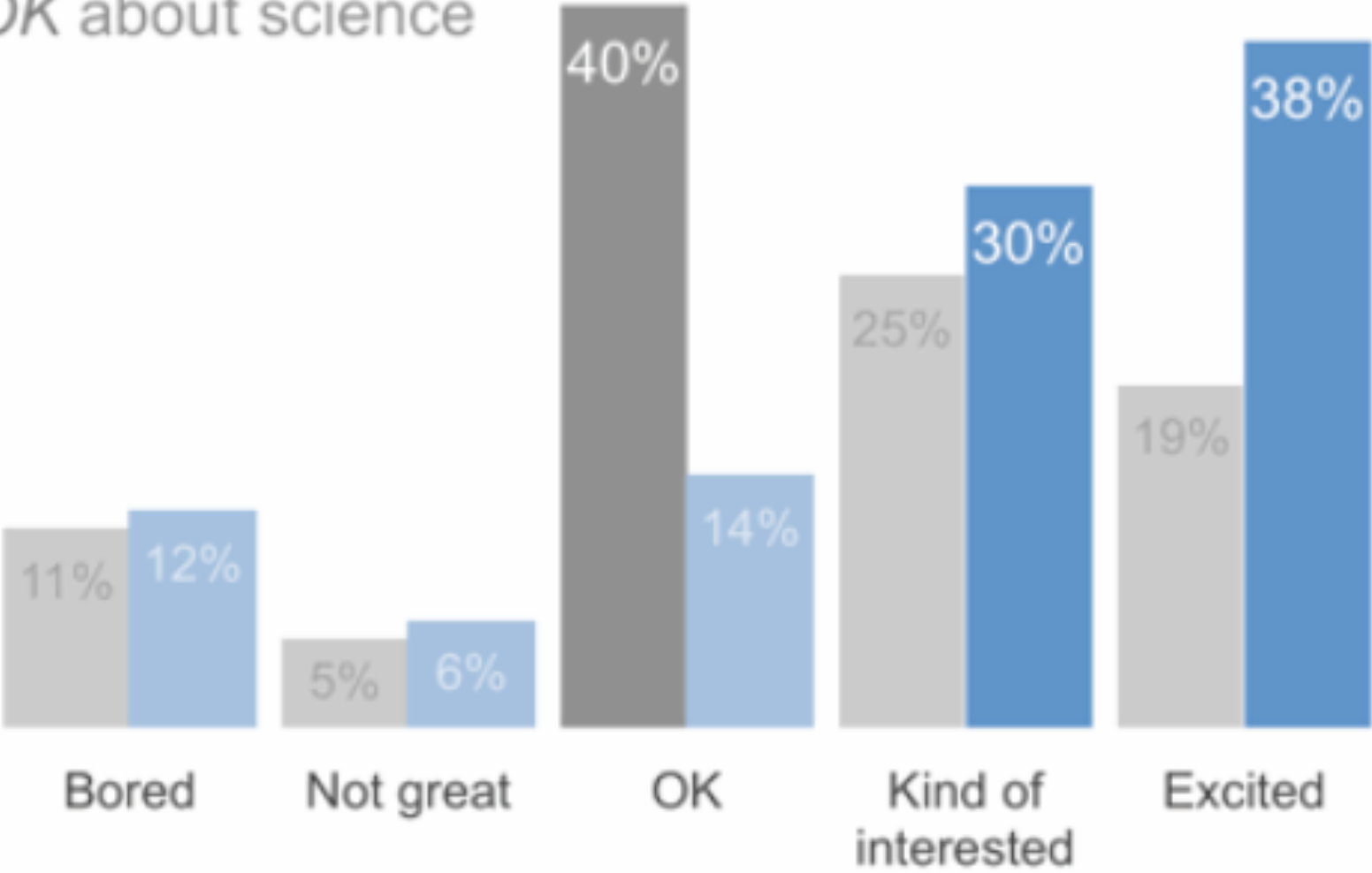
68%

of kids expressed interest towards science,
compared to 44% going into the program.

BAR GRAPH

How do you feel about science?

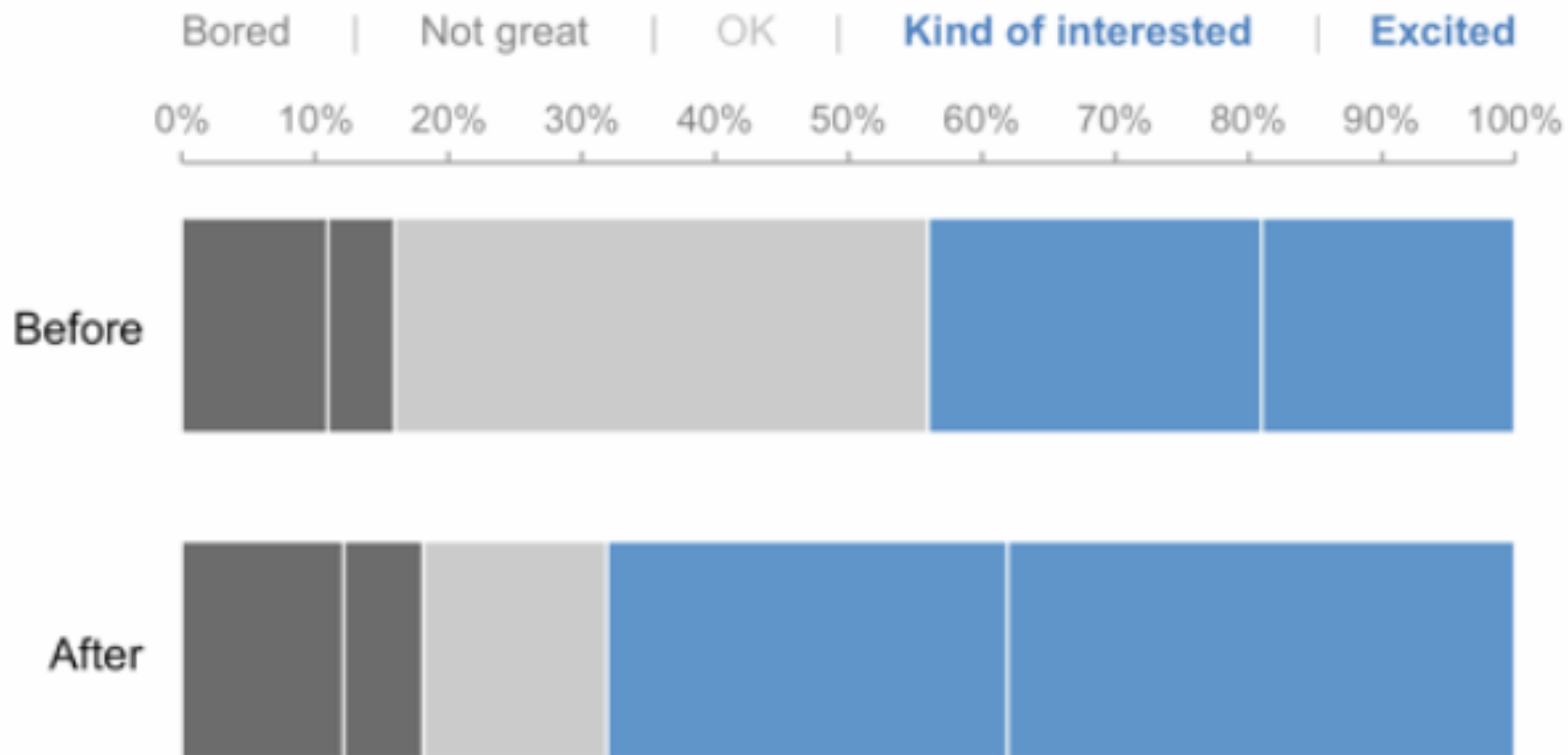
BEFORE program, the majority of children felt just *OK* about science



AFTER program, more children were *Kind of interested & Excited* about science.

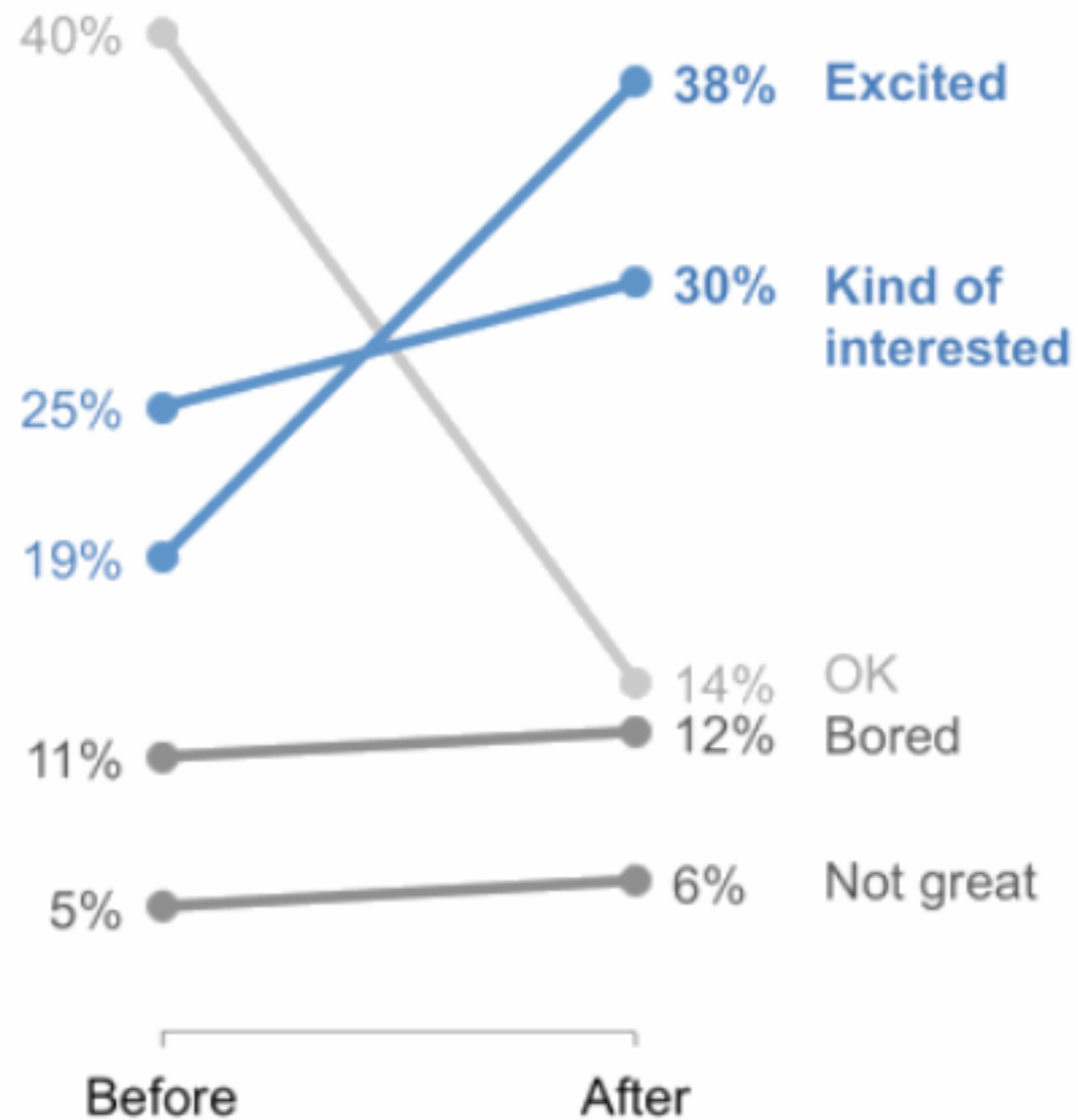
STACKED HORIZONTAL BAR GRAPH

How do you feel about science?



SLOPE GRAPH

How do you feel about science?

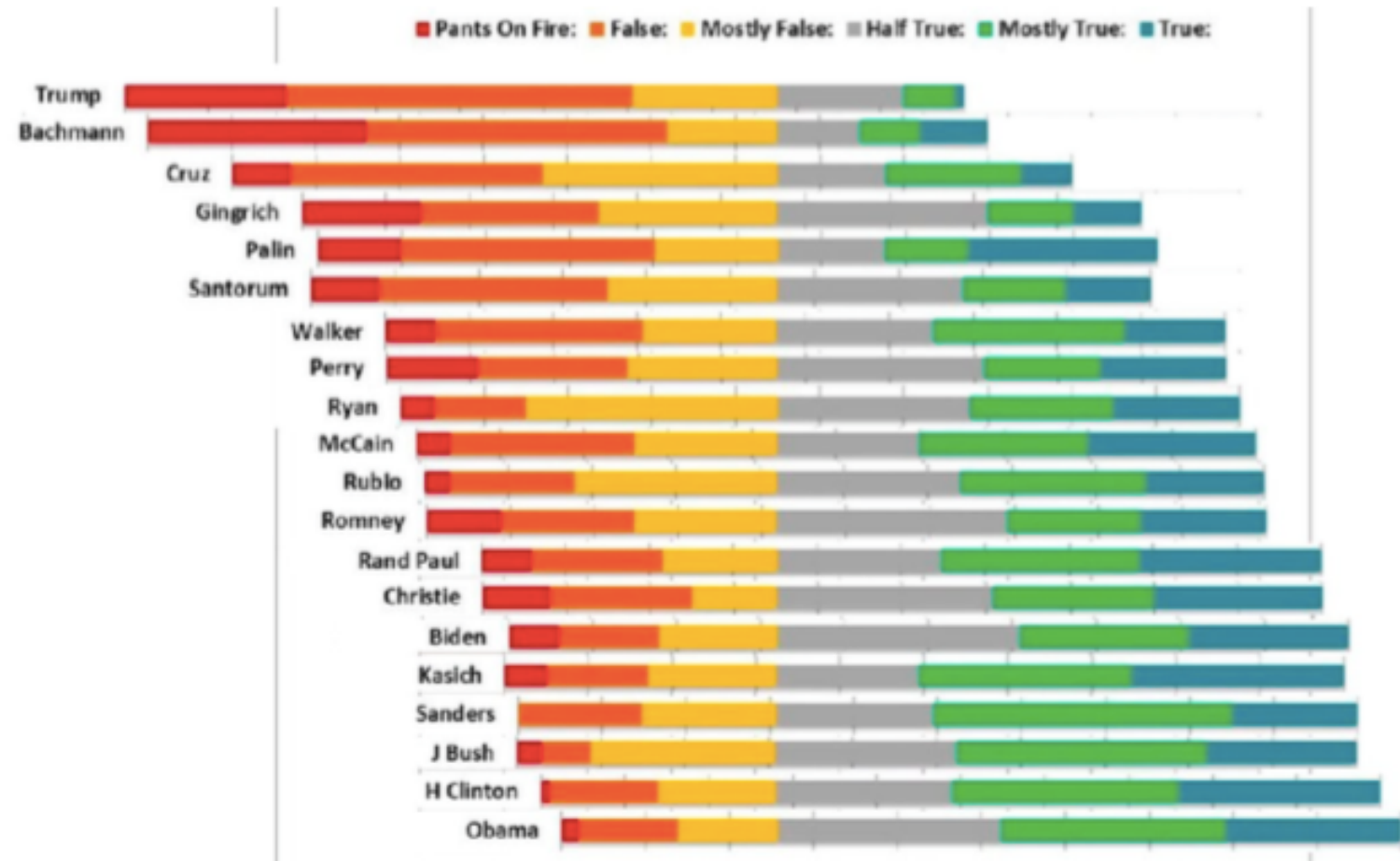
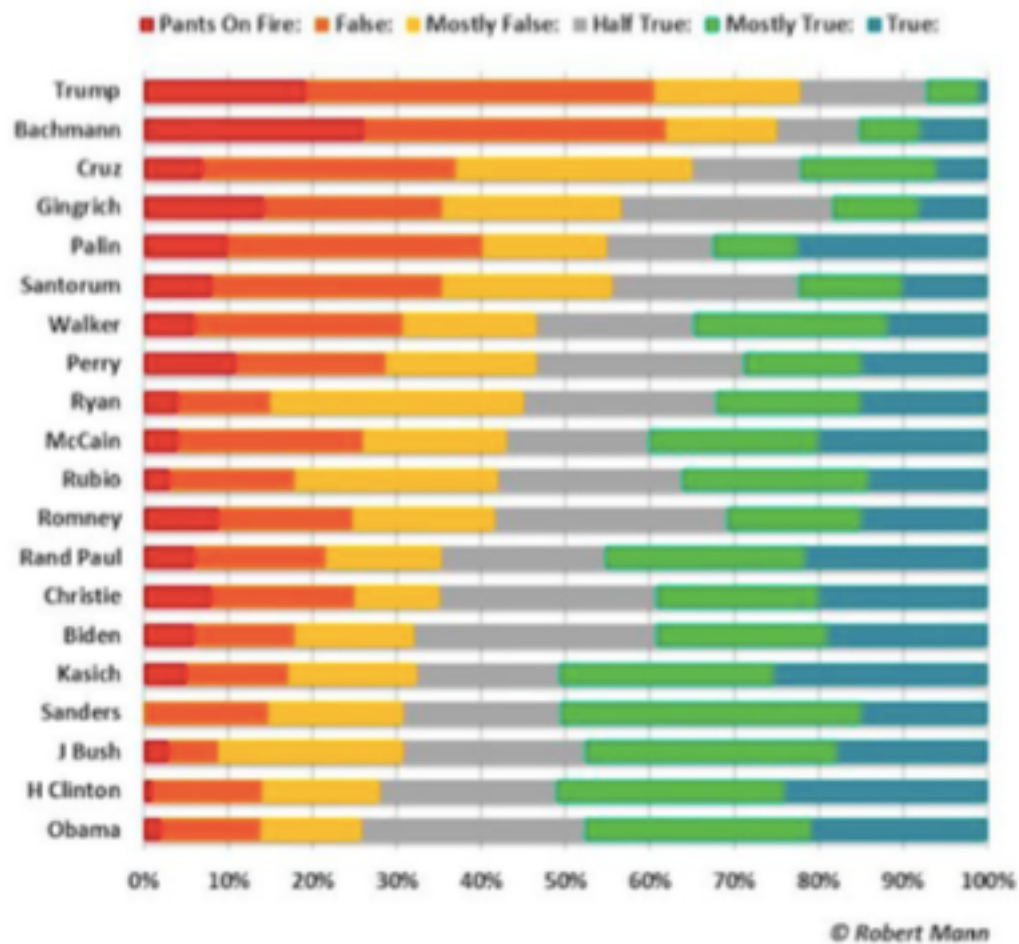


ALIGNMENT

► Stacked horizontal bar graph

Who Lies More: A Comparison

PolitiFact, an independent fact-checking website, has graded more than 50 statements since 2007 from each of these candidates. Here is how they rank.





Photographic images
used in this presentation:

mostly from Microsoft Clipart circa 2009
(unless otherwise specified)
Note references are linked to images in PDF
version of this presentation.

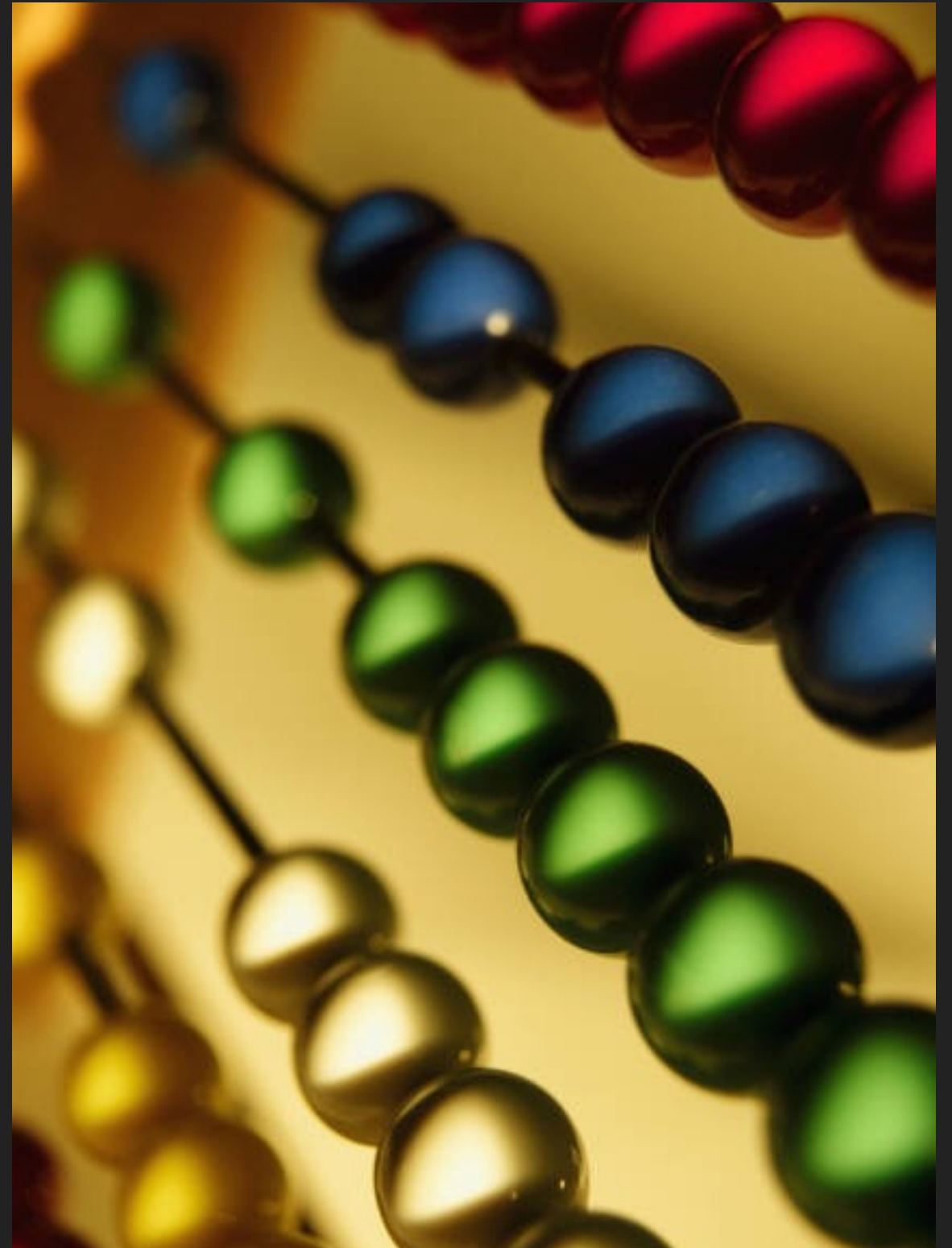
COS30045

DATA VISUALISATION

TOPIC 02-3: DESIGN GUIDELINES TUFTE'S DESIGN PRINCIPLES PART 1

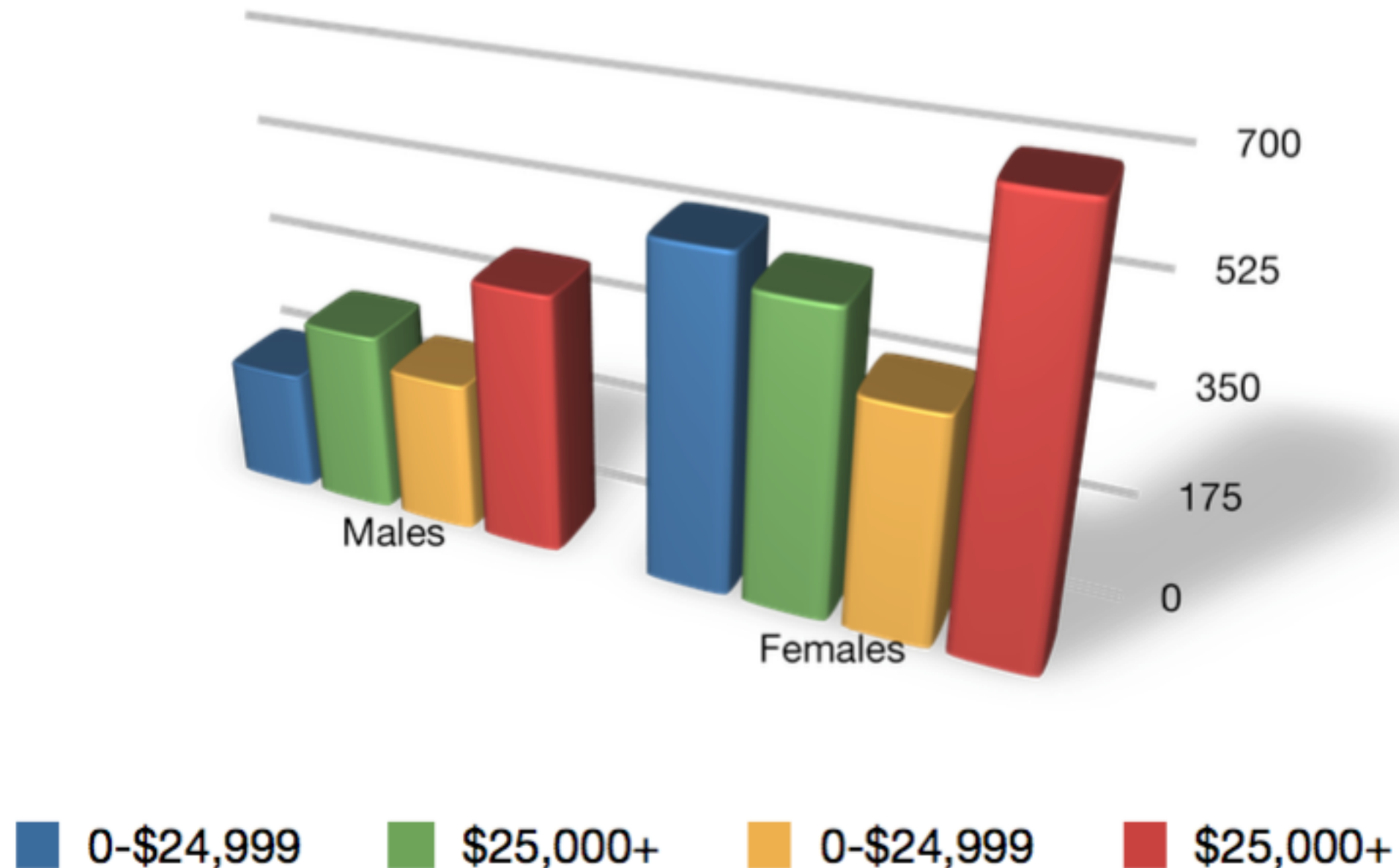
TUFTE'S DESIGN PRINCIPLES

- ▶ Maximise data-ink ratio
- ▶ Avoid chart junk
- ▶ Aim for high data density
- ▶ Layering and separation of information



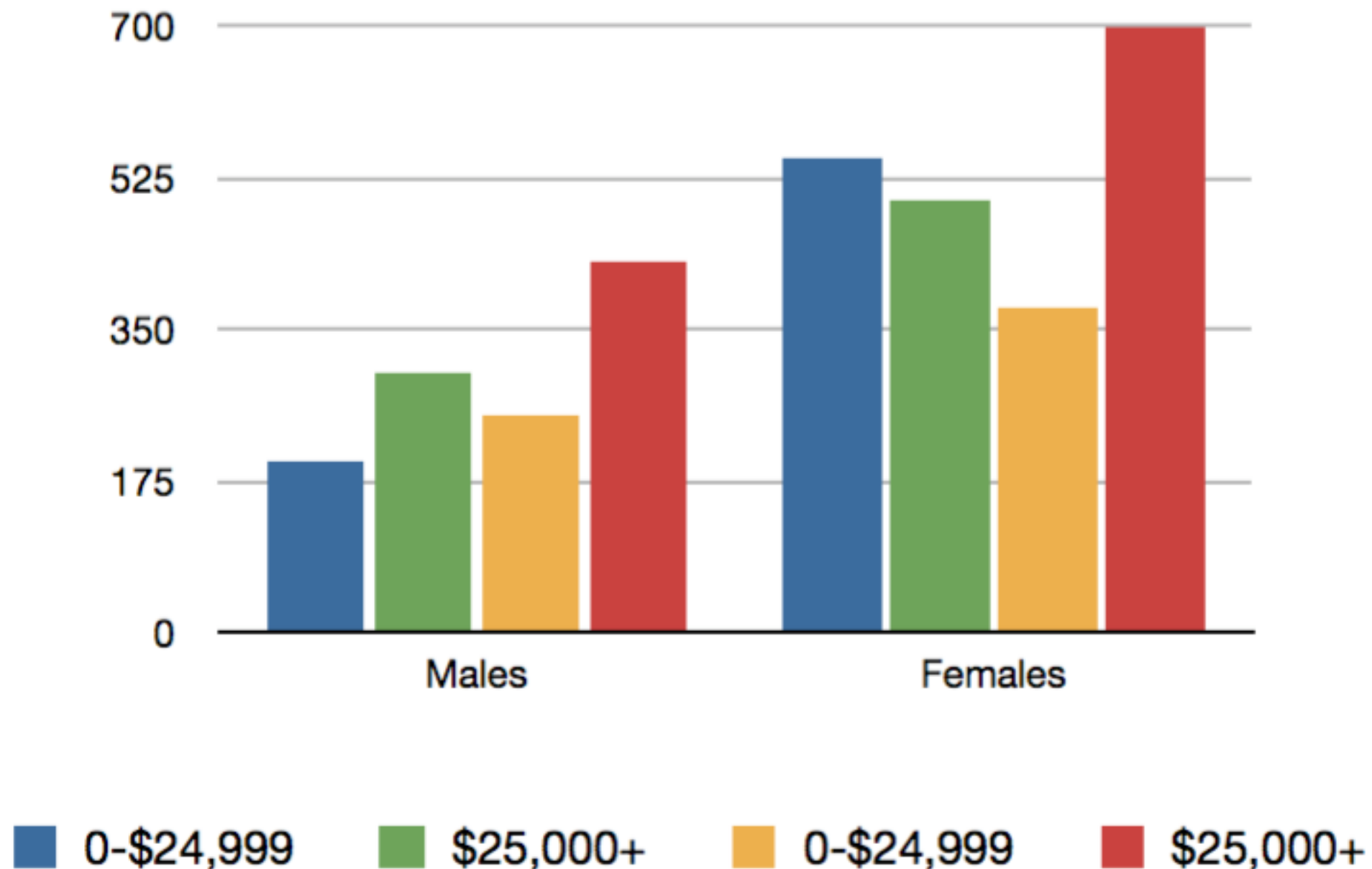
DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

$$\text{Data-Ink Ratio} = \frac{\text{data-ink}}{\text{total ink used in graphic}}$$



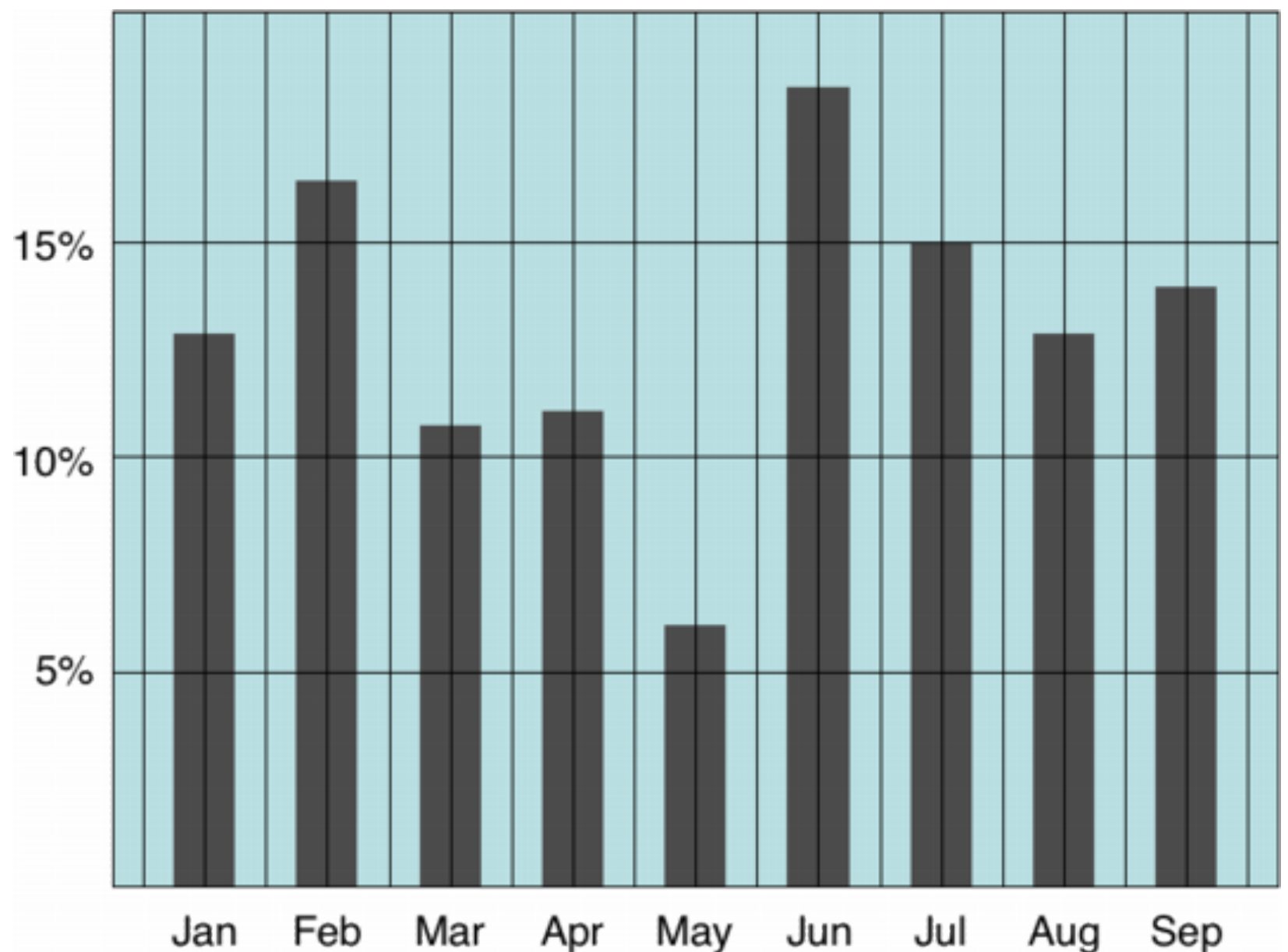
DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

$$\text{Data-Ink Ratio} = \frac{\text{data-ink}}{\text{total ink used in graphic}}$$



DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

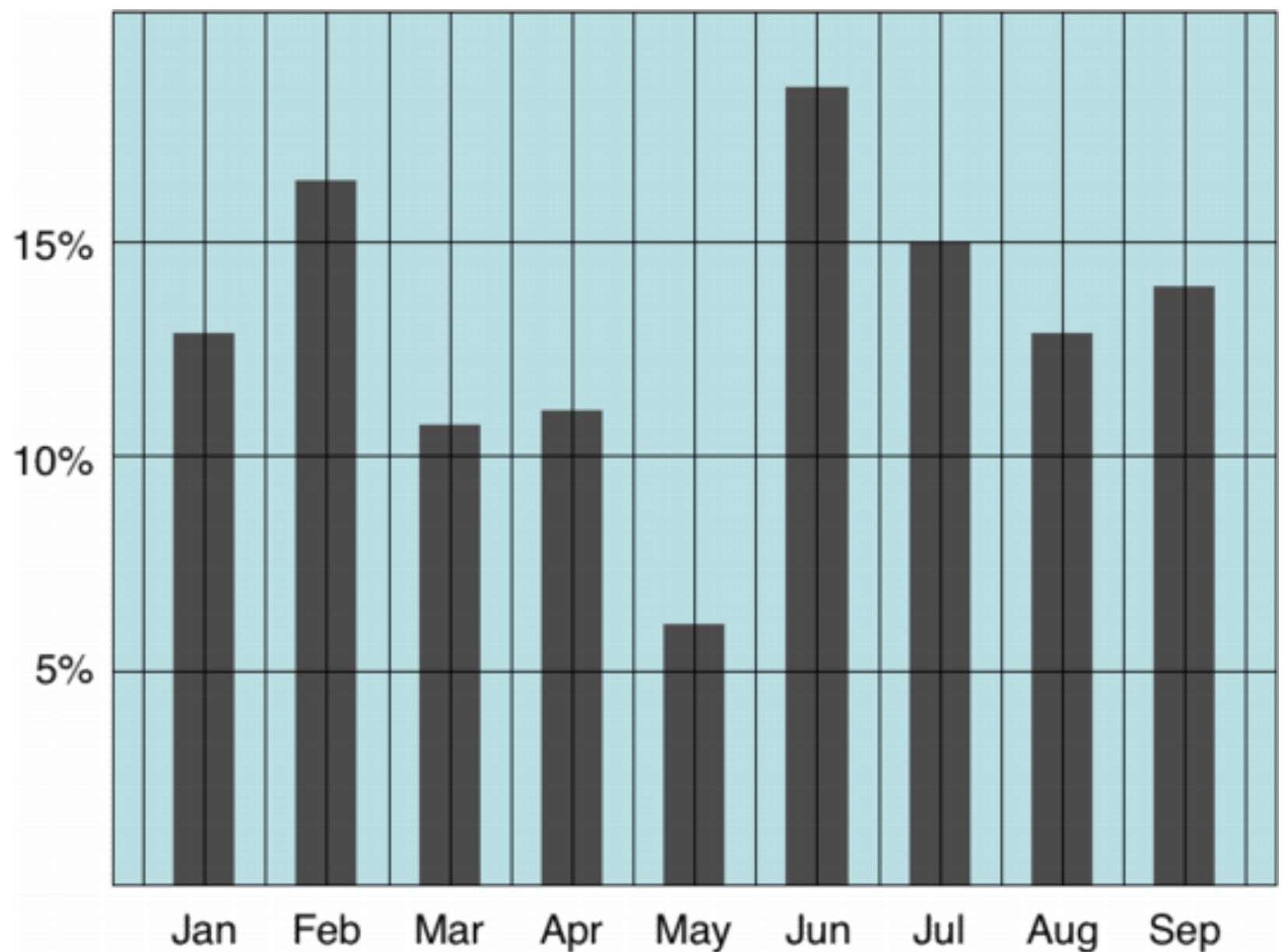
- ▶ Avoid 'Chart Junk':
 - ▶ Chart Junk: extraneous visual elements the distract from the message



DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

Bar Graph

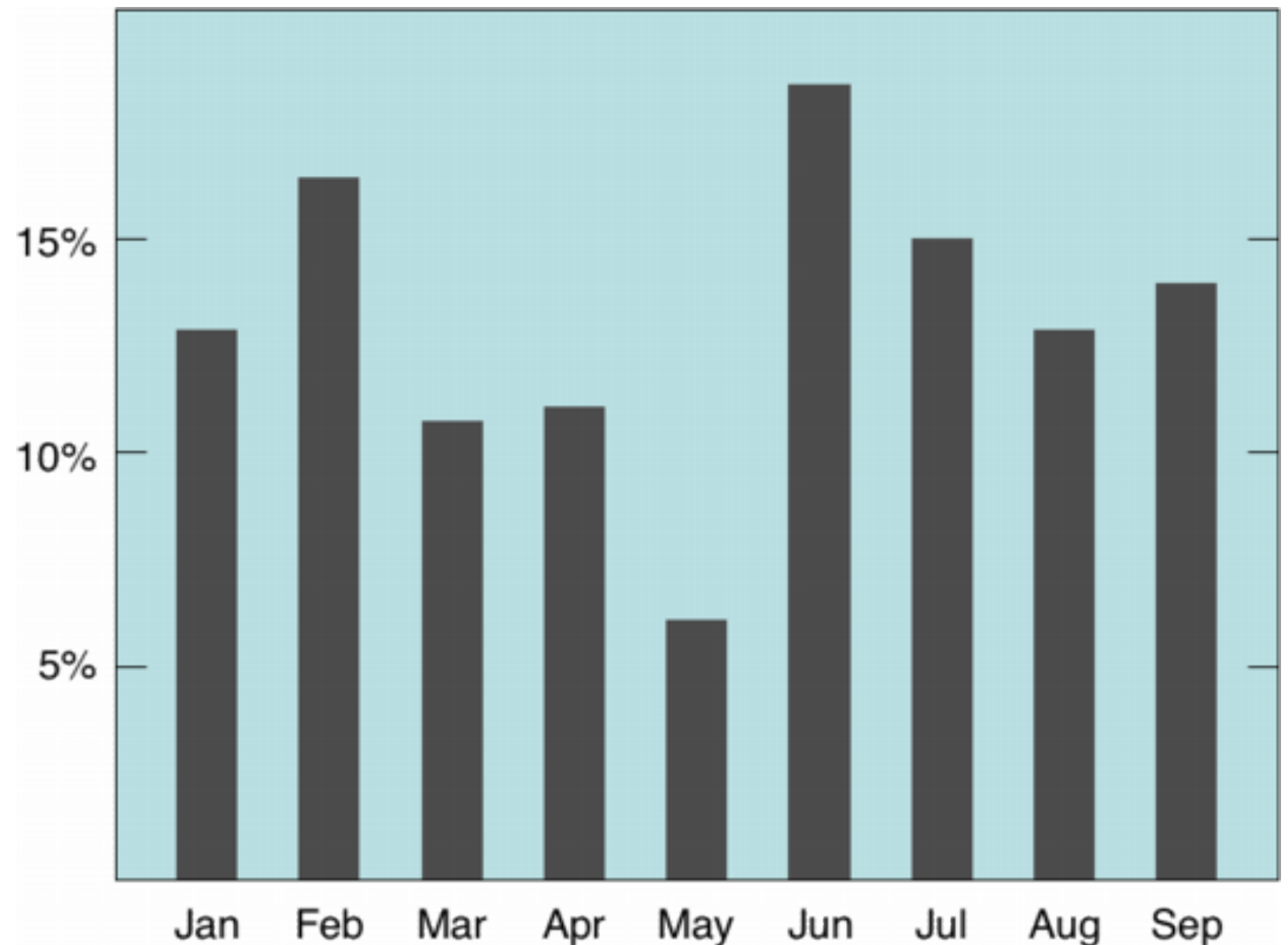
- ▶ 9 data points
- ▶ Very poor data-ink ratio



DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

Bar Graph

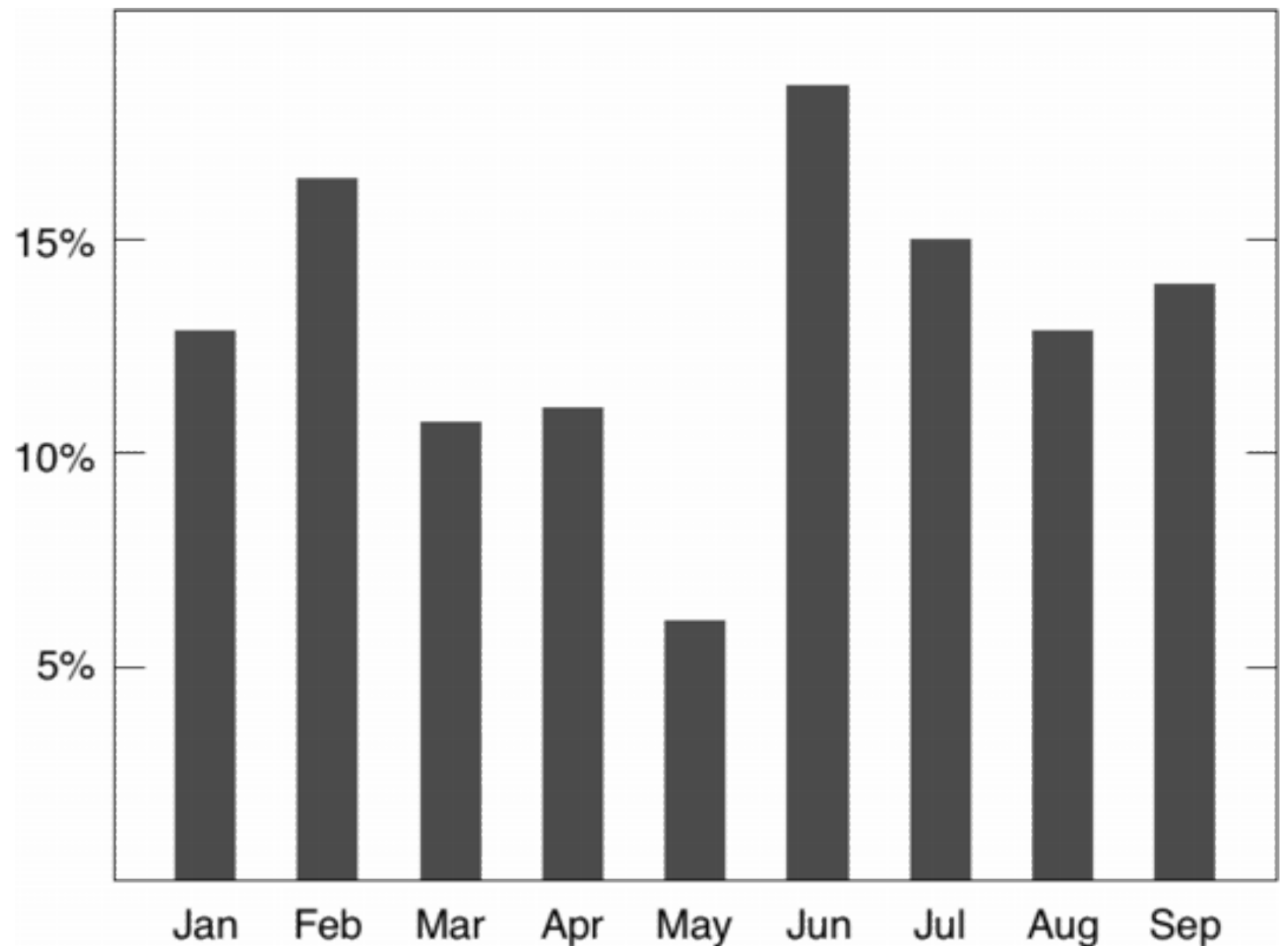
- ▶ remove the grid



DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

Bar Graph

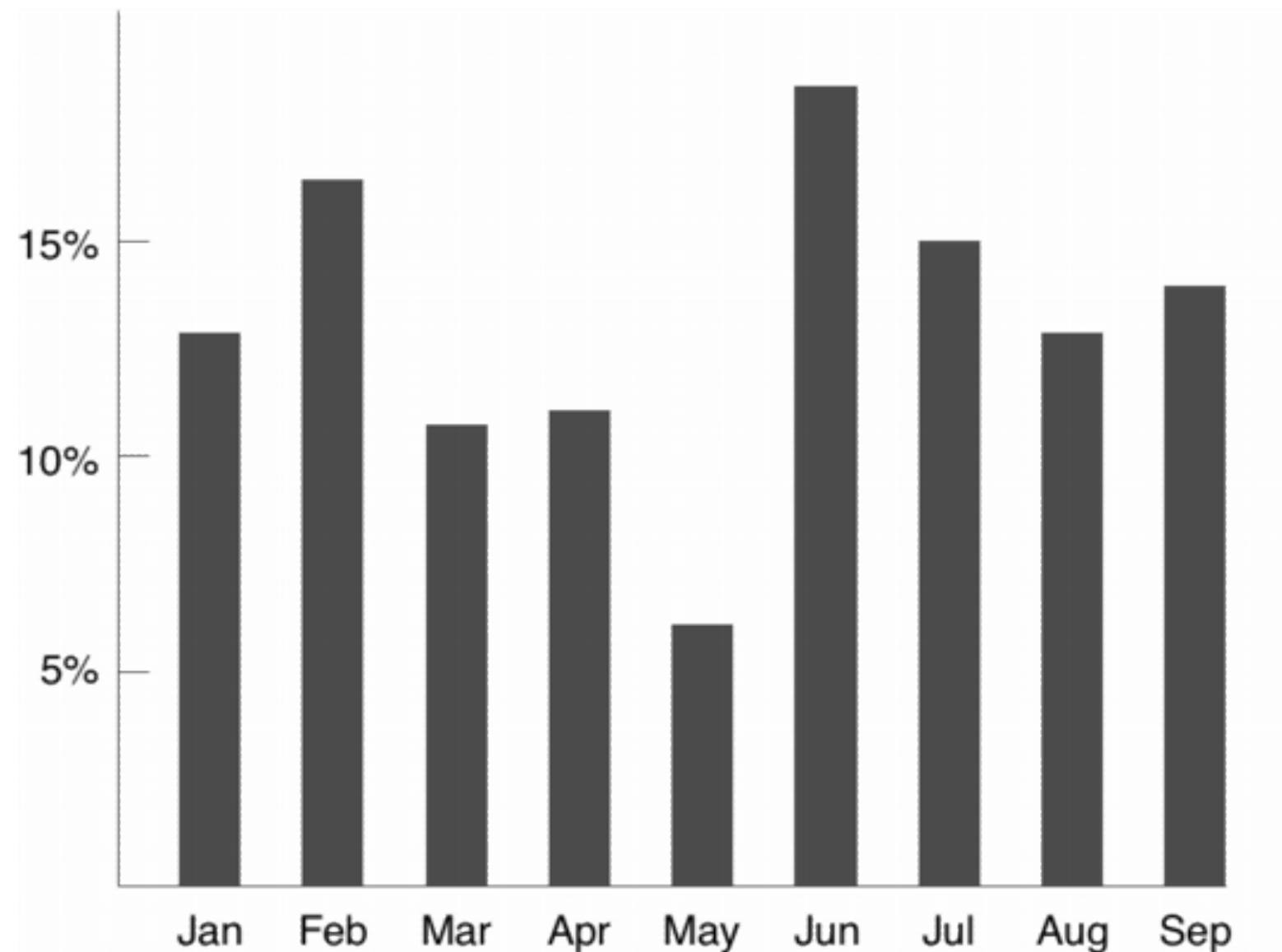
- ▶ remove the background



DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

Bar Graph

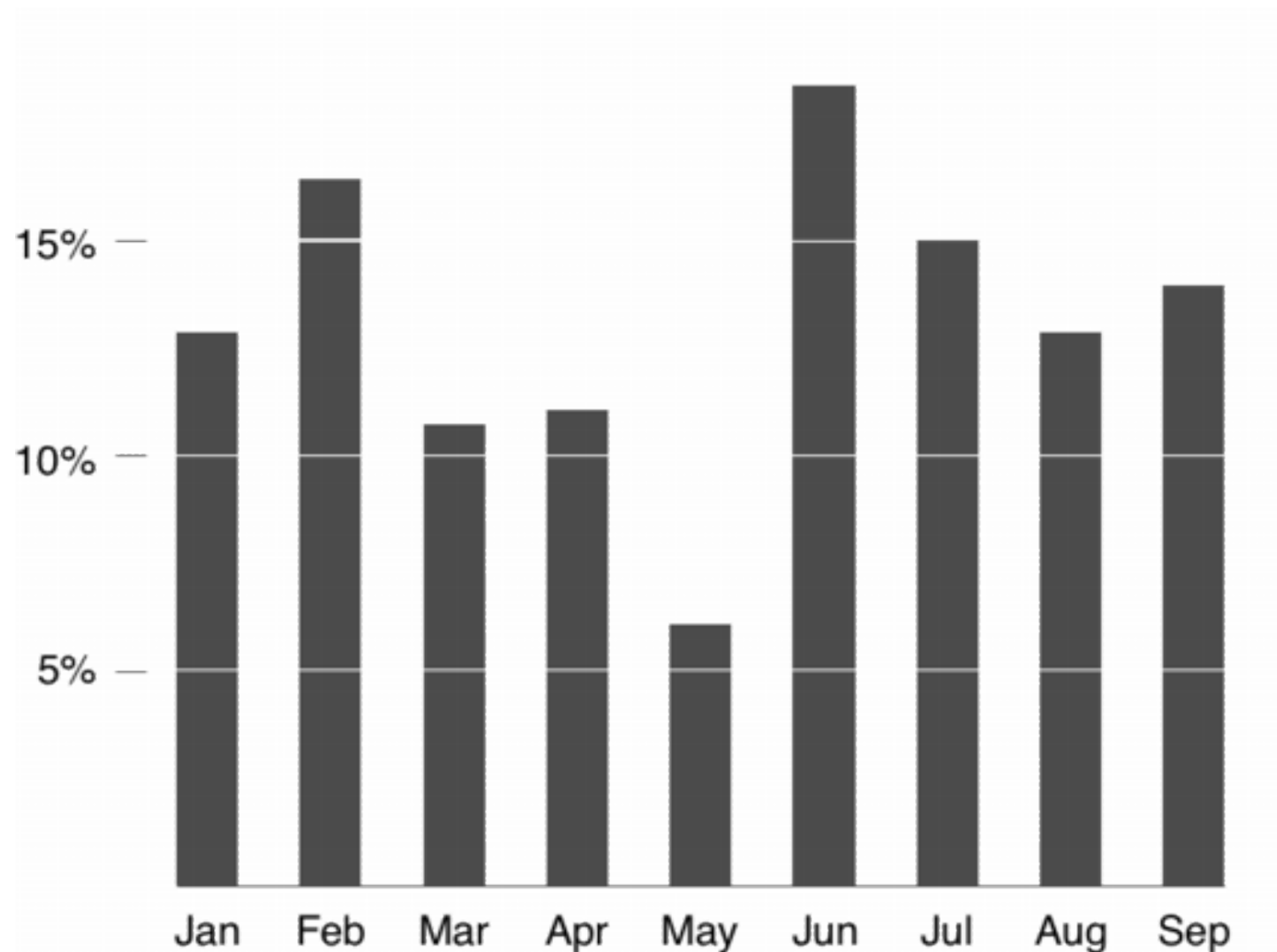
- ▶ remove the box



DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

Bar Graph

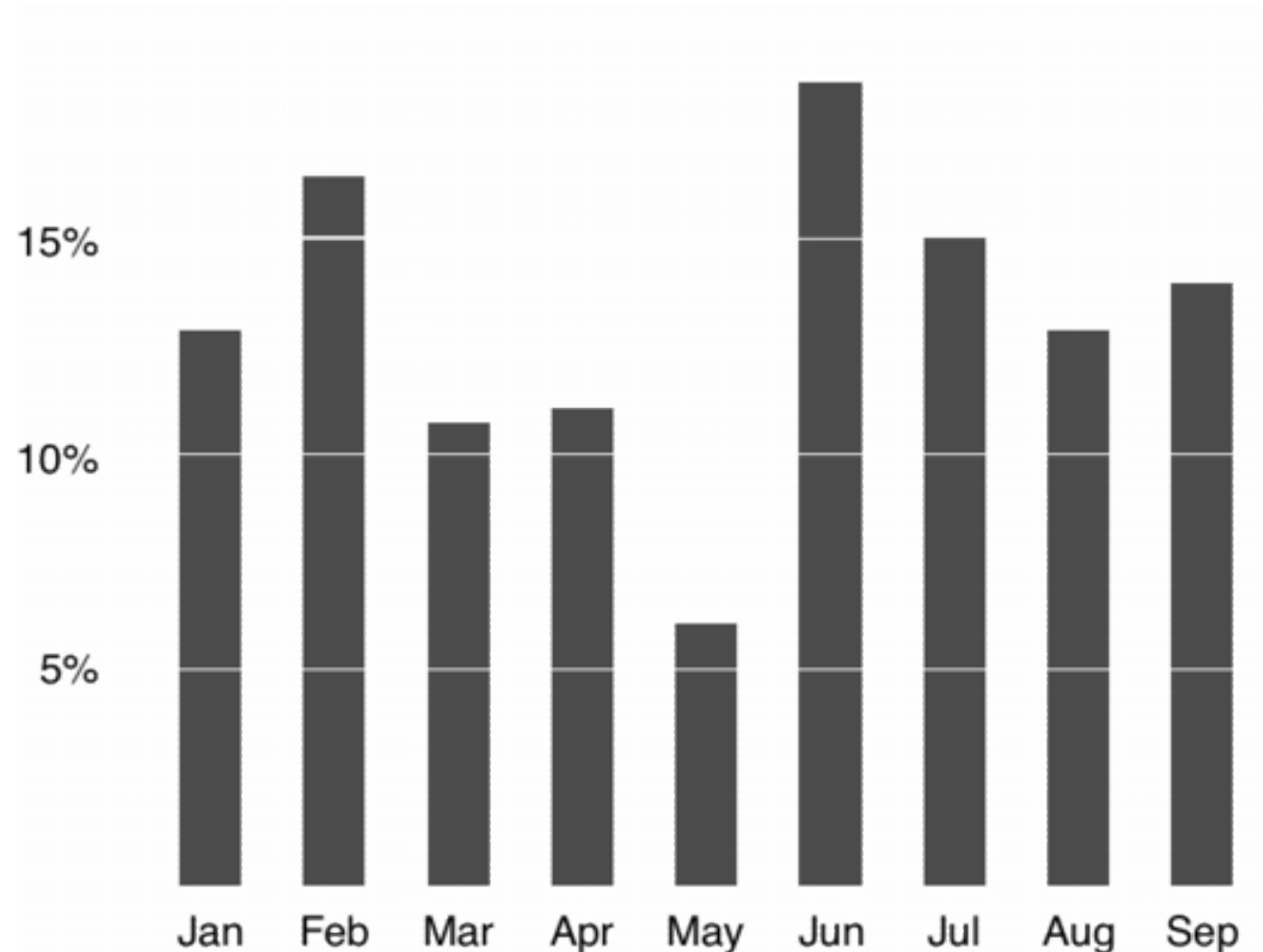
- ▶ subtract ink to make horizontal scale



DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

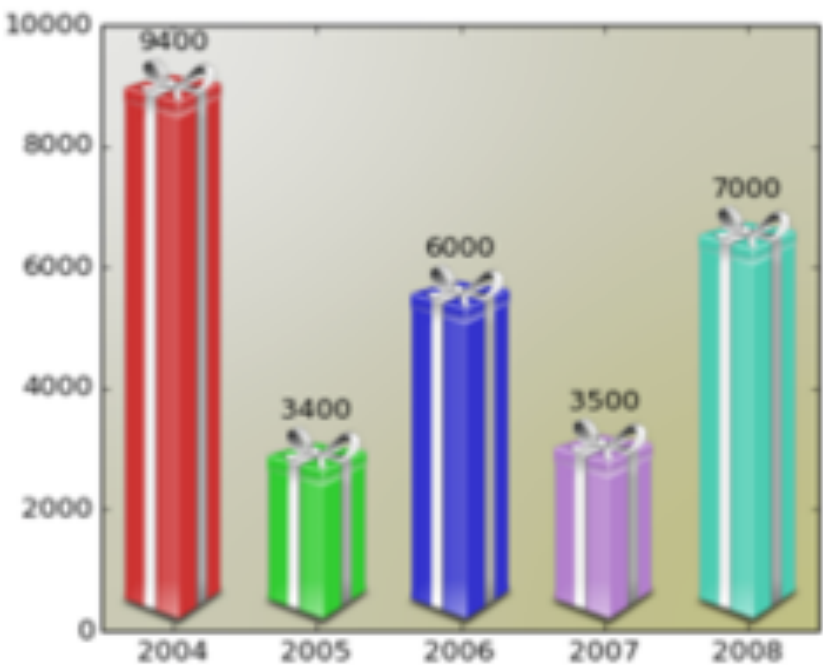
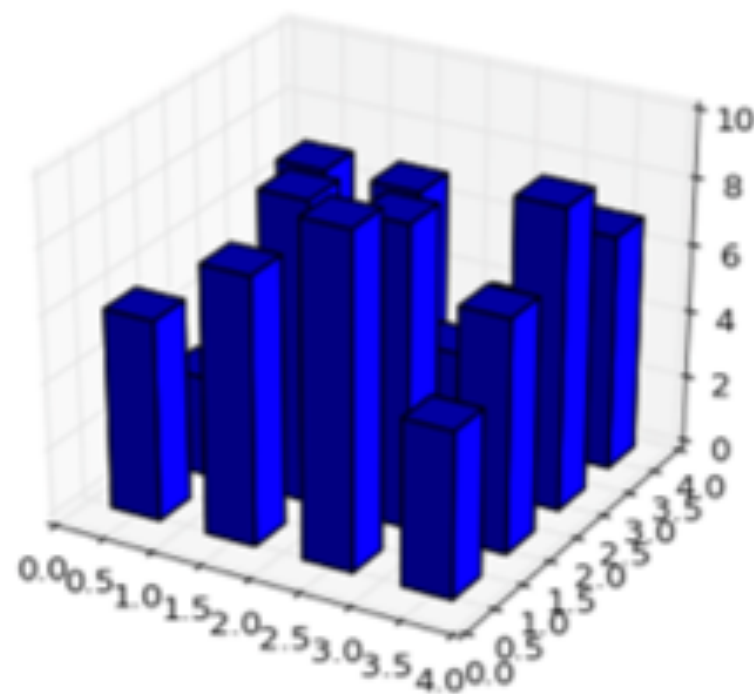
Bar Graph

- ▶ take out base line and horizontal ticks
- ▶ every pixel communicates information about the data

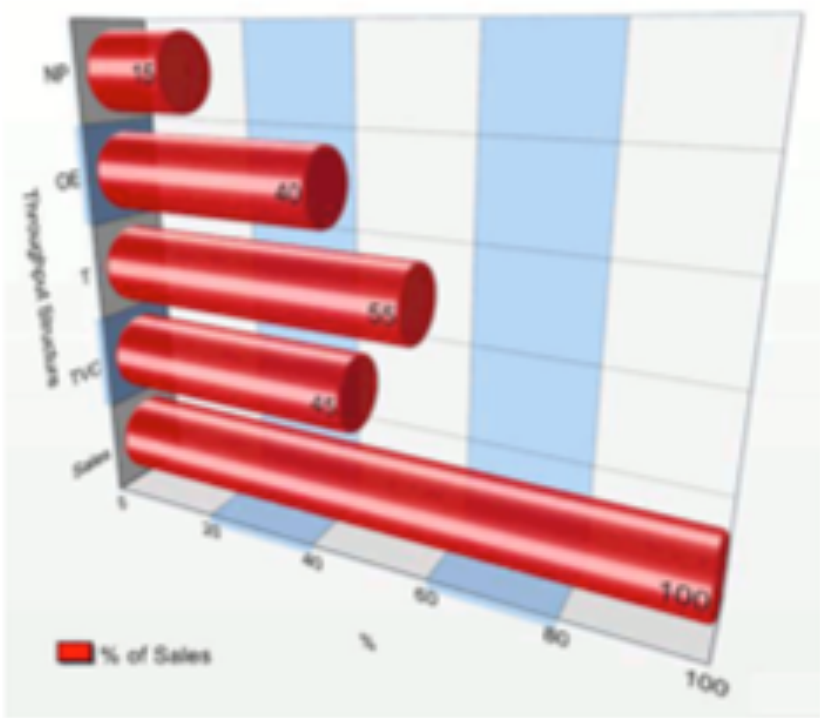


DESIGN PRINCIPLE: MAXIMIZE DATA-INK RATIO

Too much
'ink'!!

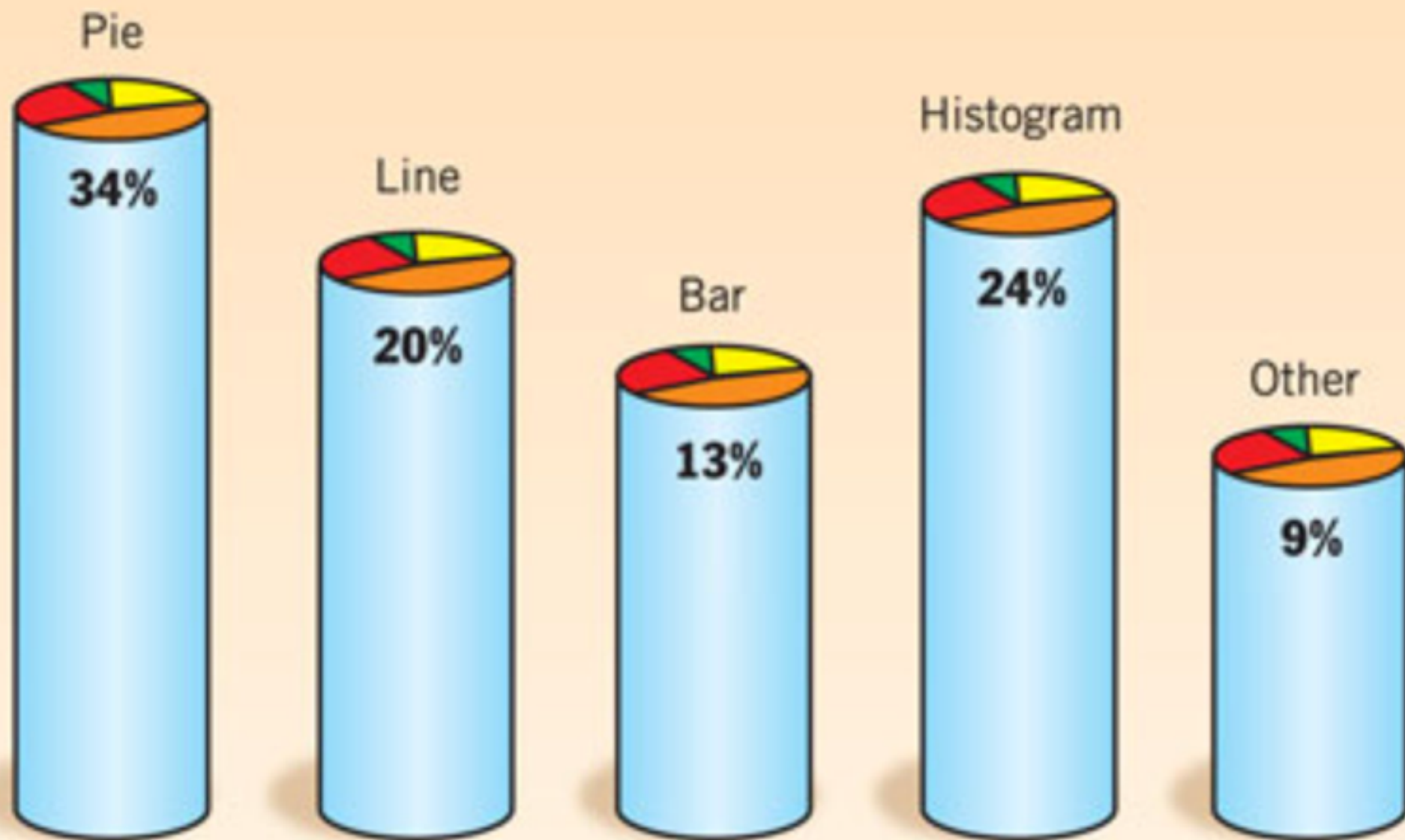


matplotlib gallery



Excel Charts Blog

America's Most Popular Charts



DESIGN PRINCIPLE: DATA DENSITY

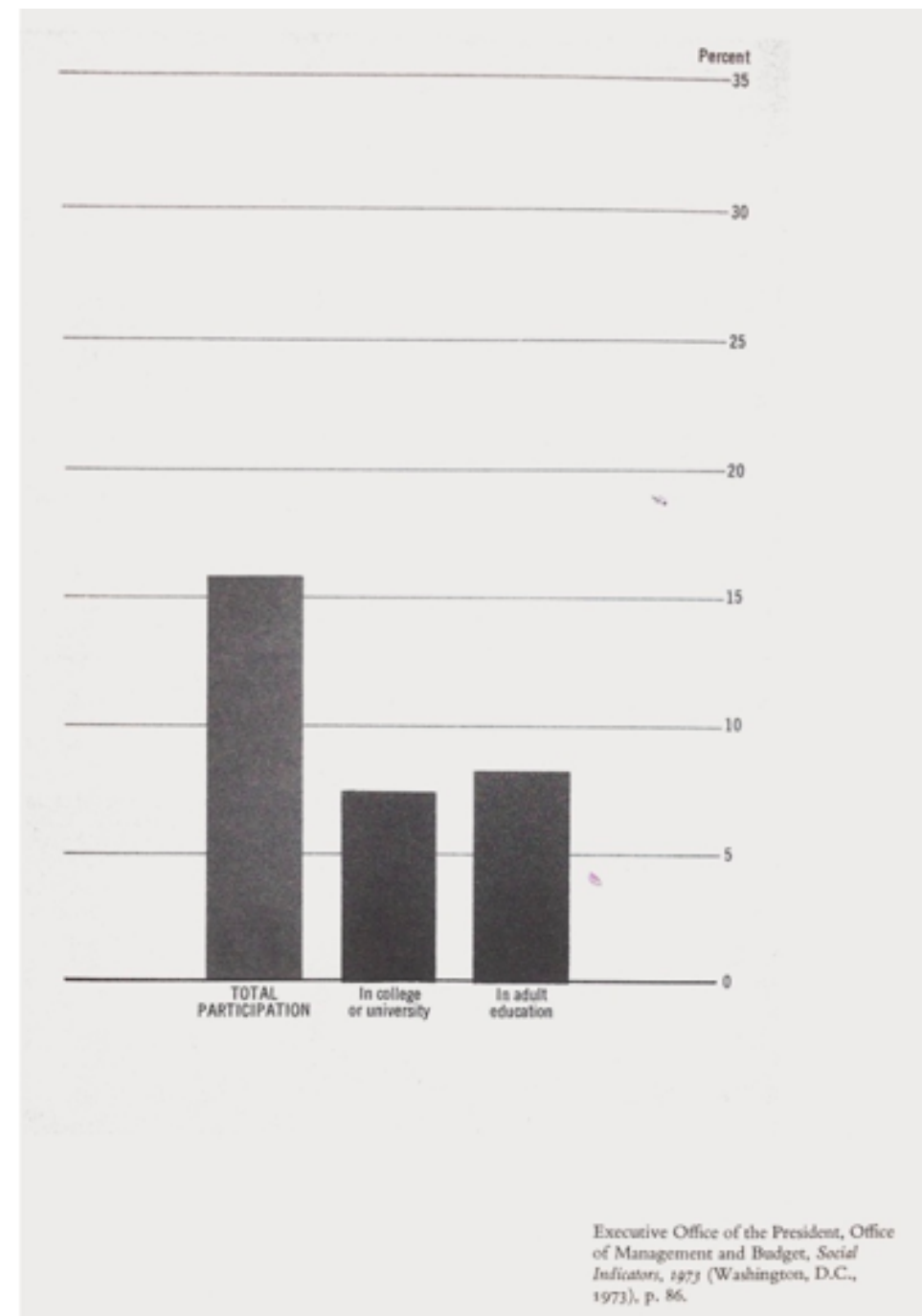
Data Density = $\frac{\text{number of data items}}{\text{area of data in graphic}}$

- ▶ aim for high data density

6-POINT	BELL GOTHIC
Vaught Donald L 542 39th St Short Wylam	780-8608
Vaught Ernest	491-6244
Vaught Joe Jr 625 Barclay Ln	836-2436
Vaught Ralph L 700 77th Wy S	836-8452
Vaught Susan A 2109 46th Pl Central Pl	787-4227
Vaultz Eva 1543 Dennison Av SW	925-1752
Vause S F 603 Hickberry Ln	979-5289
Vause Stephen F 445 Shades Crest Rd	823-2662
Vautier Harold G 204 Killough Springs Rd	853-5626
Vautrot Ruby L Mrs 2021 10th Av S	933-2265
Vazquez Norberto	674-3370
Veach J L 9128 Belmont Dr	956-3990
Veach Loren Adrich	665-1831
Veal Ad 450 21st Av S	251-9049
Veal Ad 1711 Pinson	841-7380
Veal B Evanatty 1711 Pinson	841-2789
Veal Clarence E Garndie	631-3856
VEAL CONVENTION SERVICES	
1711 Pinson	841-2789
2109 10th Av N	322-6102
Veazey W B Vincent	872-9556
Veazey Wilbur E 1541 53rd St Enley	923-1960
Veazey William A 287 -A Chastaine Cir	942-4137
Veazey Willie J 3084 Whispering Pines Cir	823-5795
Veber Mark H 5216 Goldmar Dr	956-1661
Veeco contr 1900 28th Av S Homewood	879-2259
Vedel Dental Technicians Inc Lab	
1116 5th Av N	322-5475
Vedel George C 3848 Cromwell Dr	967-2832
Vedel George C Jr 744 Sauter Ln	871-8234
Res 34744 Sauter Ln	870-9758
Vedel Murray B 612 Oakmoor Dr	942-3619
Vedell Colleen J Daisy City	774-7772
Vedell William L 8830 Valley Hill Dr	833-9915
Veenschoten & Co mfrs agts 2930 7th Av S	251-3567
Veenschoten L A 1818 -D Tree Top Ln	822-7109
Veenschoten W E 3240 Pine Ridge Rd	871-8883
Vega Abraham 915 16th S	933-7619
Vega Delores 2 -B Waterman Cir	836-5980
Vega Edwin 2116 Rockland Dr Bluff Park	823-0403
Vegetable Patch Number 1 The	
Highway 31 S Alabstr	663-7618
Vegetable Patch Office Alabstr	663-7378
Vegetable Patch The Number 2 Dogwood	665-1179
Veig Patrick B Pawnee	841-1238
Veitch Beulah 1172 Five Mile Rd	853-3361
Vest W L 4708 Lewisburg Rd	841-7402
Vest W T 4737 N 66th	836-6371
Vesta Villa Exxon Self Serve	823-5008
500 Hwy 31 S	823-5008
VESTAVIA AMOCO SERVICE	
1456 Montgomery Hwy	823-1213
VESTAVIA BARBEQUE & LOUNGE	
610 Montgomery Hwy Vestavia	822-9984
Vestavia Barber Shop	
610 -A Montgomery Hwy	823-1974
VESTAVIA BEAUTY SALON	
710 Montgomery Hwy	823-1893
Vestavia Beverage Co	
623 Montgomery Hwy	822-9847
VESTAVIA BOWL	
Montgomery Hwy S Vestavia	979-4420
Vestavia Church Of Christ	
2325 Columbiana Rd	822-0018
VESTAVIA CHURCH OF GOD	
2575 Columbiana Rd	823-1895
Vestavia Church Of God Day Care day	
nursry 2575 Columbiana Rd	823-1895
VESTAVIA CITY OF---See Vestavia Hills City Of	
VESTAVIA COIFFEURS	
617 Montgomery Hwy Vestavia	823-1104
Vestavia Country Club	
Shades Mountain	823-2451
Golf Shop Shades Mountain	822-8300
On Mondays & Before 8:30 AM Dial	
As Follows---	
Stable Shades Mountain	823-2451
Accounting Dept Shades Mountain	823-2979
Golf Course Supt Shades Mountain	823-2019
Tennis Shop Shades Mountain	823-2139
General Manager's Ofc	823-2139
Building Maintenance Shop	823-2349
Swimming Pool Shades Mounta	822-2559
Vestavia Country Club Employee's	
Lounge Shades Mountain	822-9840
Vestavia Hardware & Home Supply	

Telephone book:
very high data
density

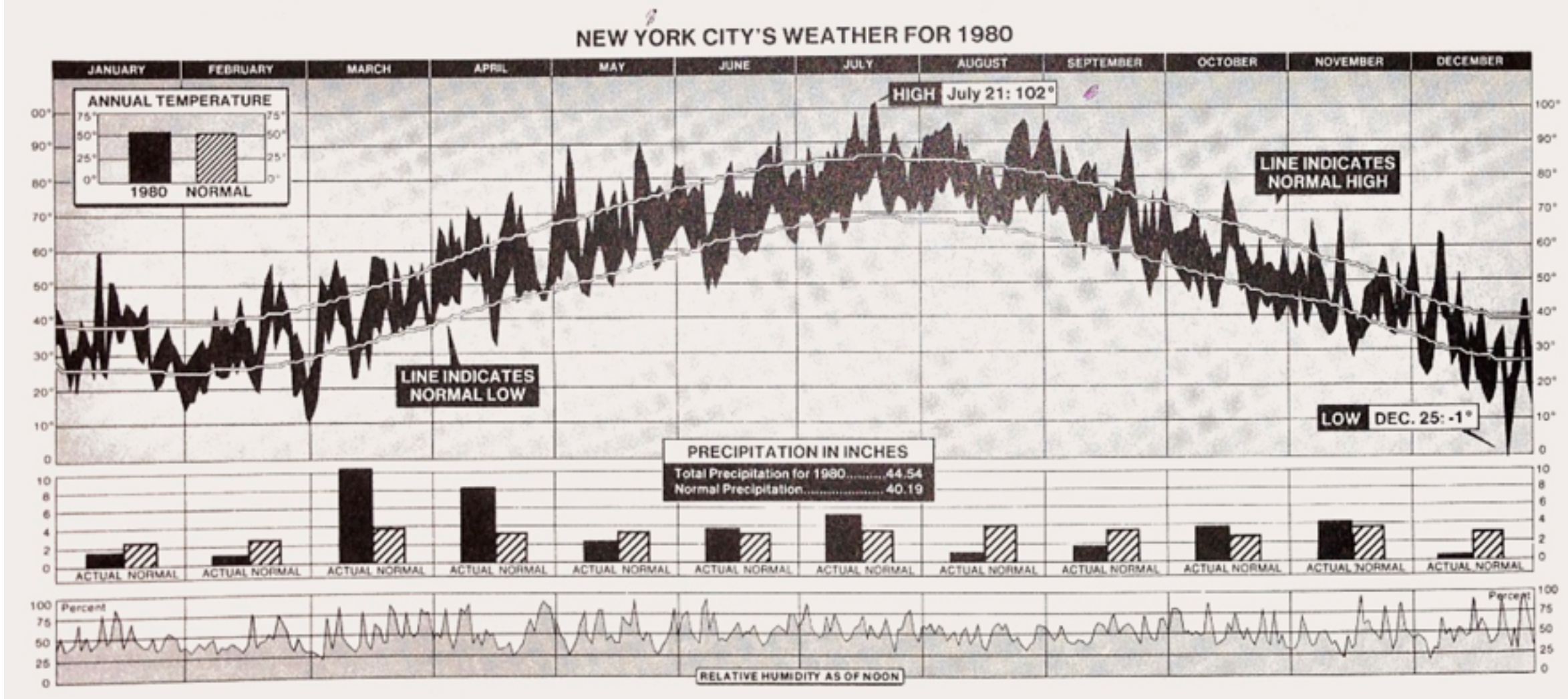
Tufte



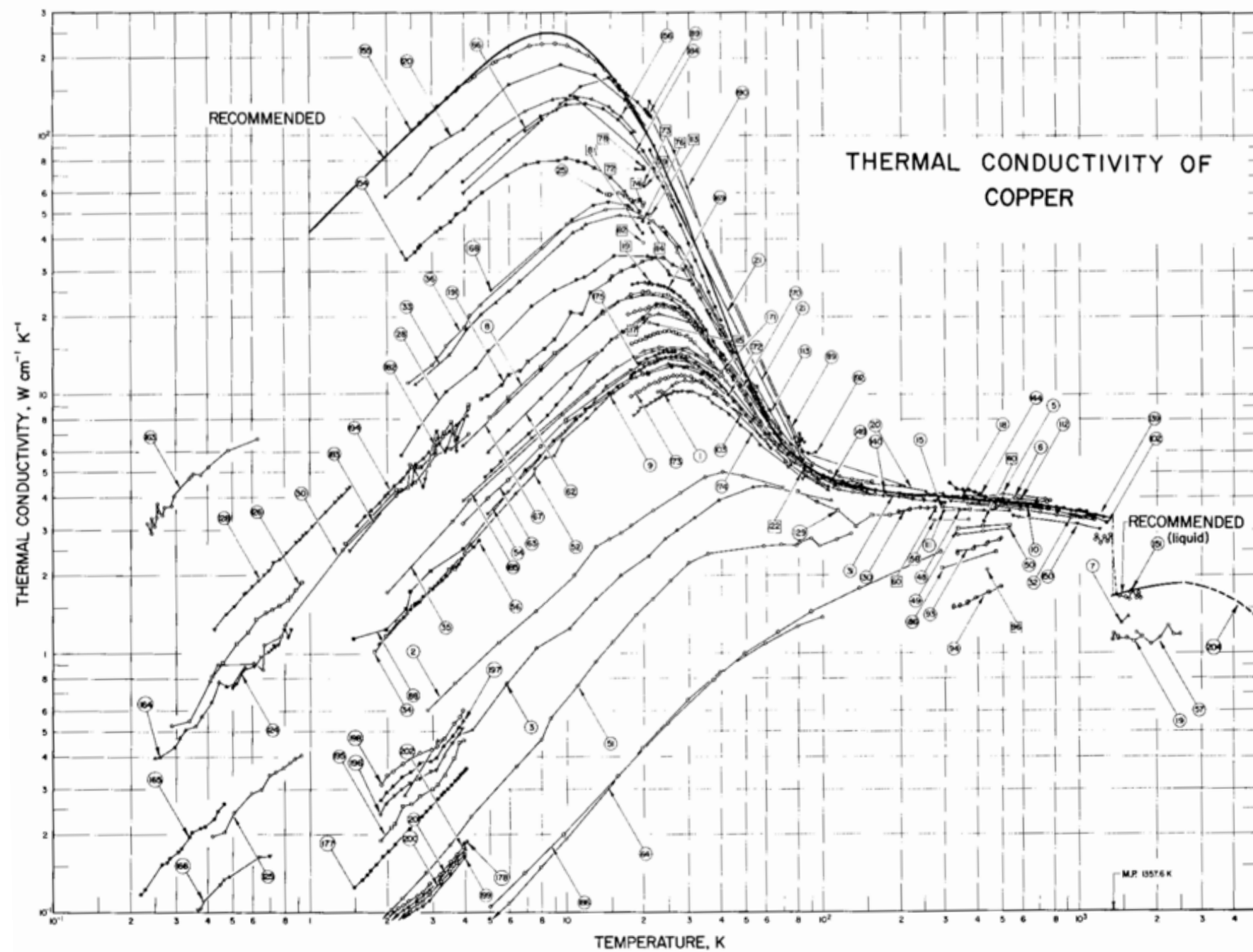
"Data matrix of 4 entries, the names and the numbers of the two bars on the right. The left bar is merely the total of the two. The graph covers 26.5 square inches, resulting in a data density of .15 numbers per square inch, which is thin indeed."

DESIGN PRINCIPLE: HIGH DATA DENSITY

In contrast, the New York weather history, in this reduced version, does very well at 181 numbers per square inch (28 per square centimeter):

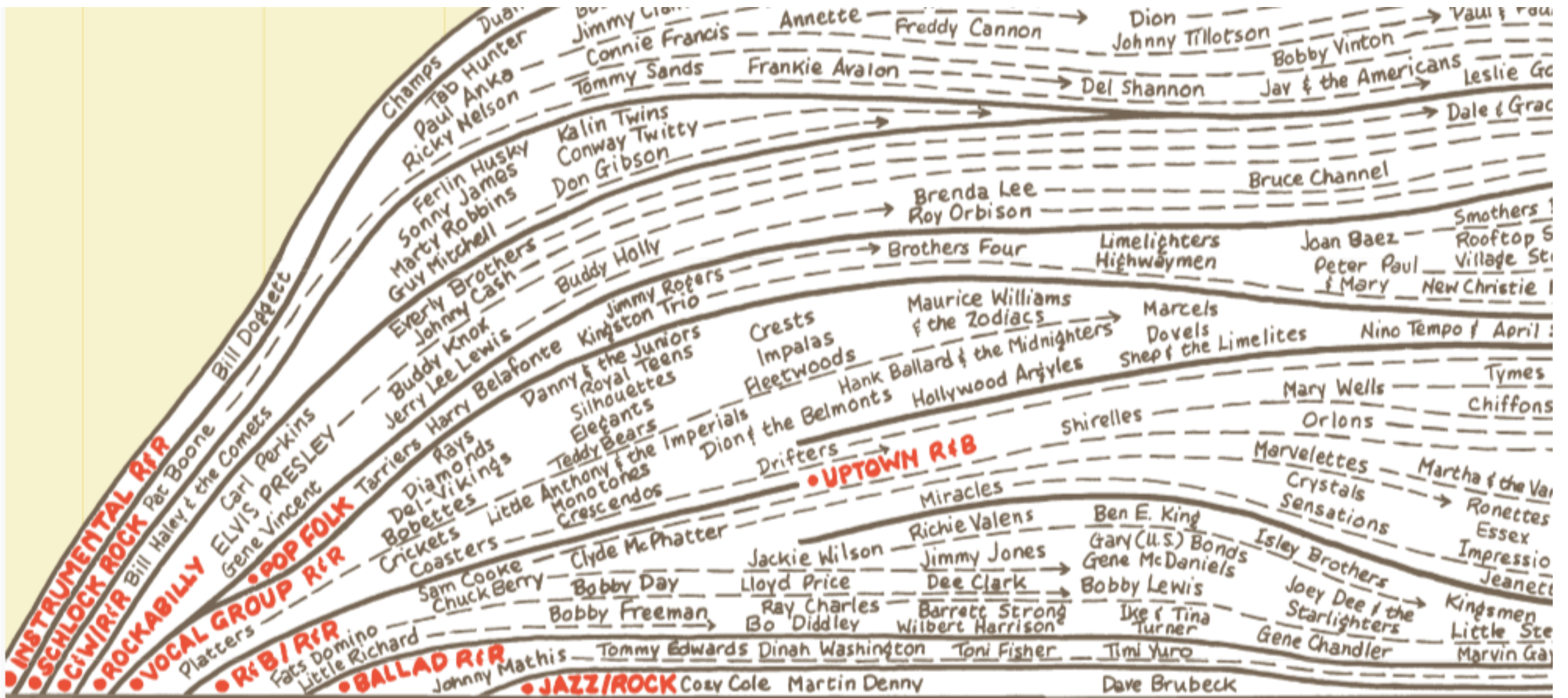


DESIGN PRINCIPLE: DATA DENSITY



Ho et al., "Thermal Conductivity of the Elements: A Comprehensive Review" J Phys. Chem. 1974

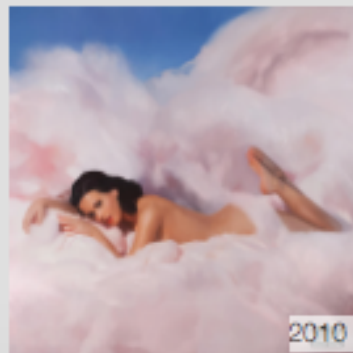
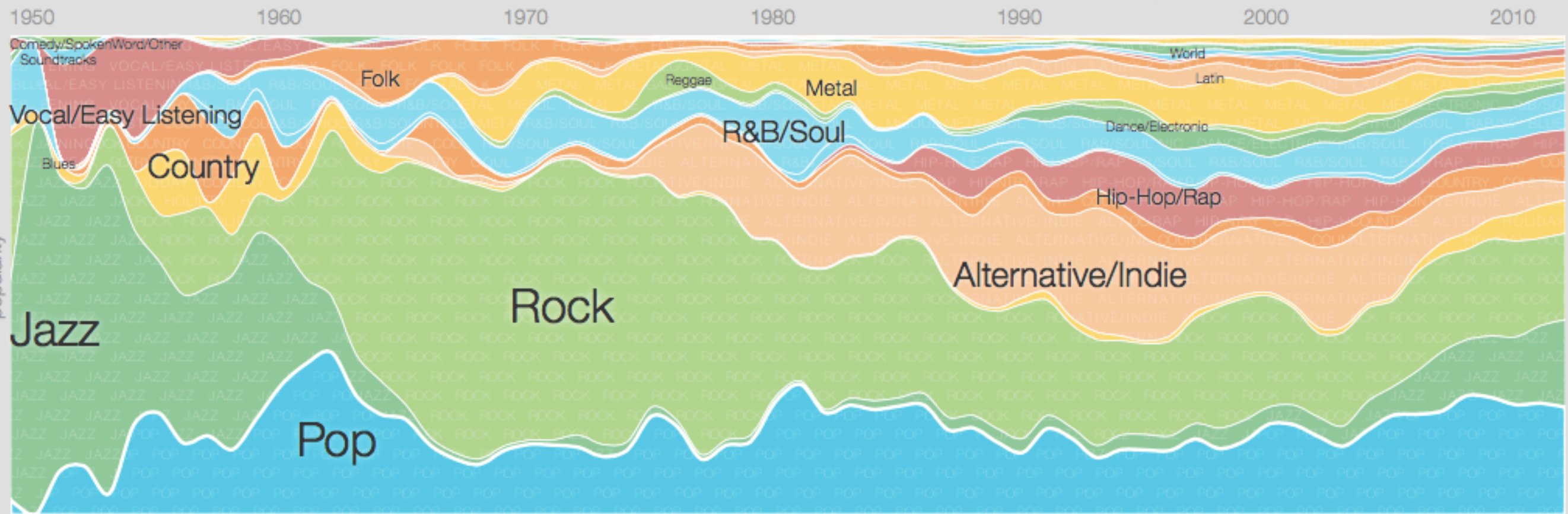
GENEALOGY OF POP/ROCK MUSIC



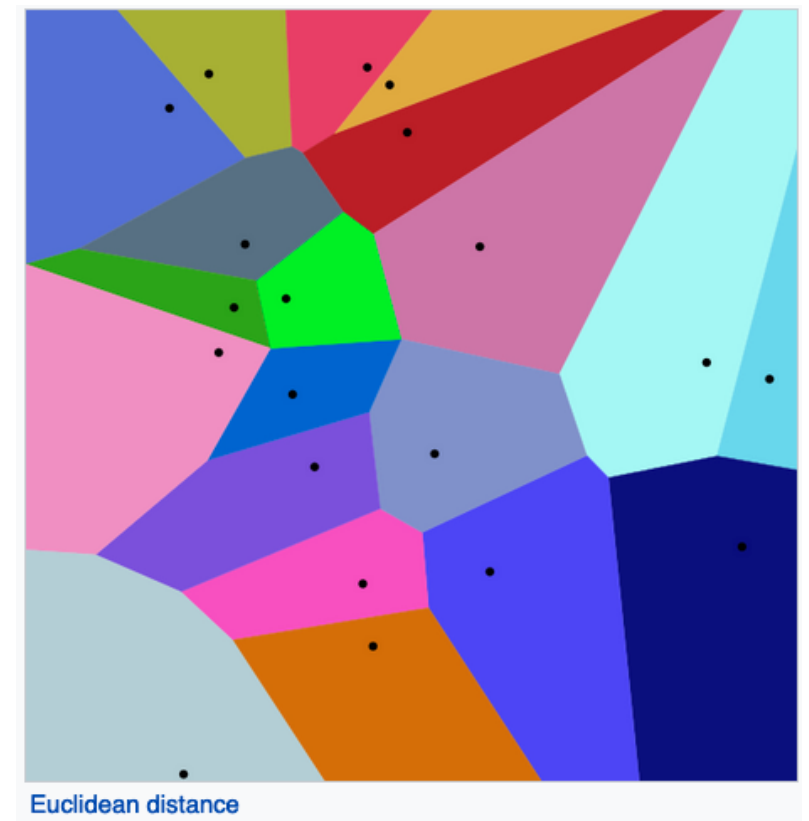
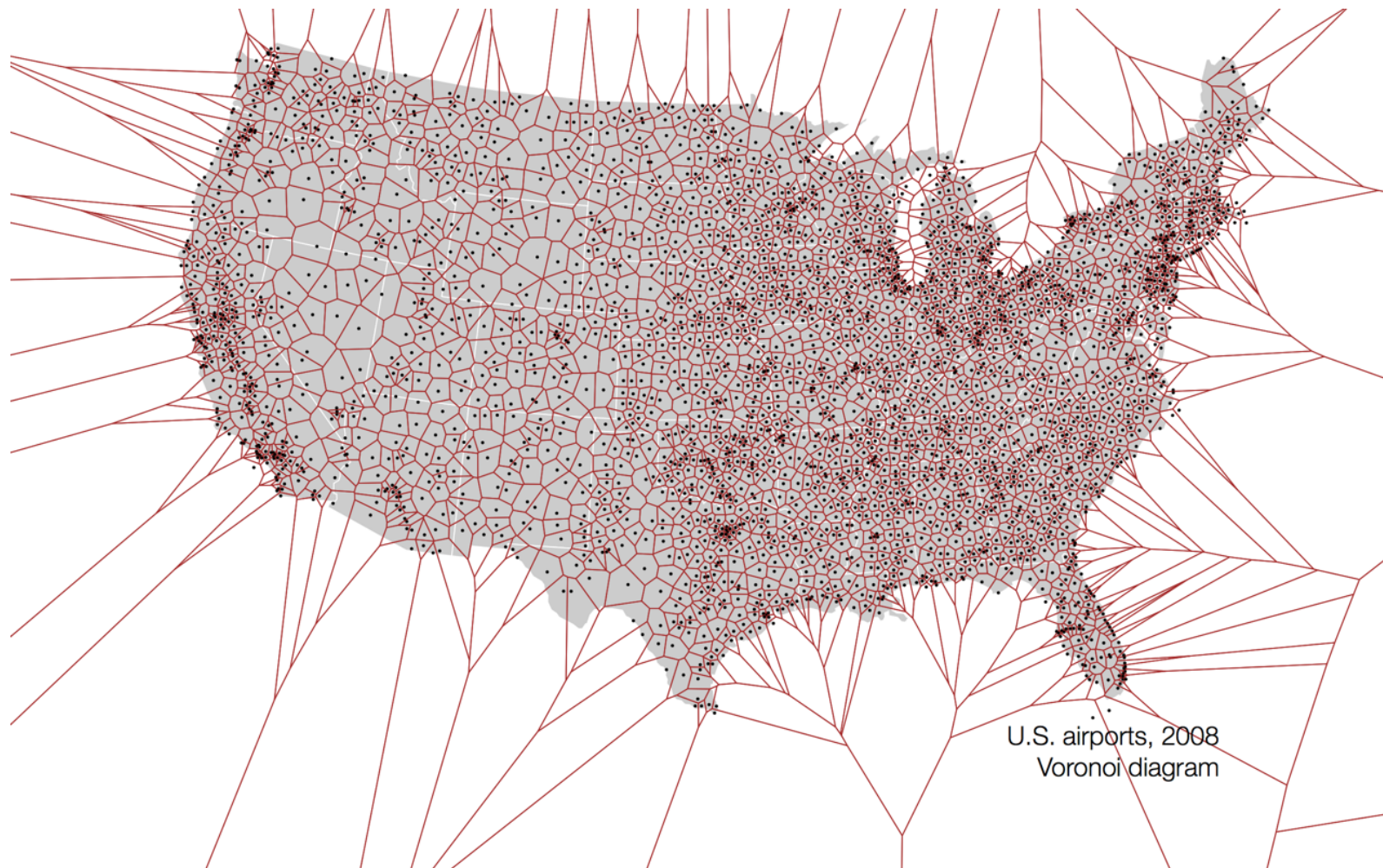
Music Timeline

Album or artist:

[FAQ](#)



DESIGN PRINCIPLE: DATA DENSITY



DESIGN PRINCIPLE: LAYERING AND SEPARATION

- ▶ Separate and layer information

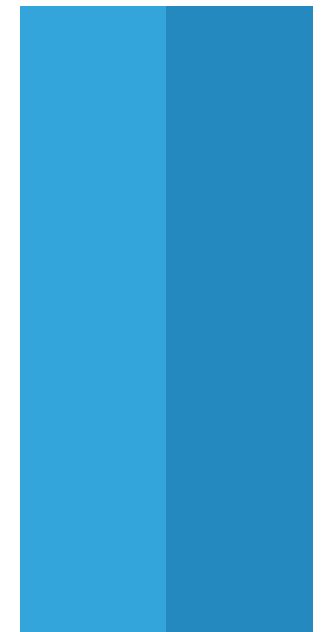
1+1=2



+



=



1+1=3

potential for
negative space to
create clutter



+



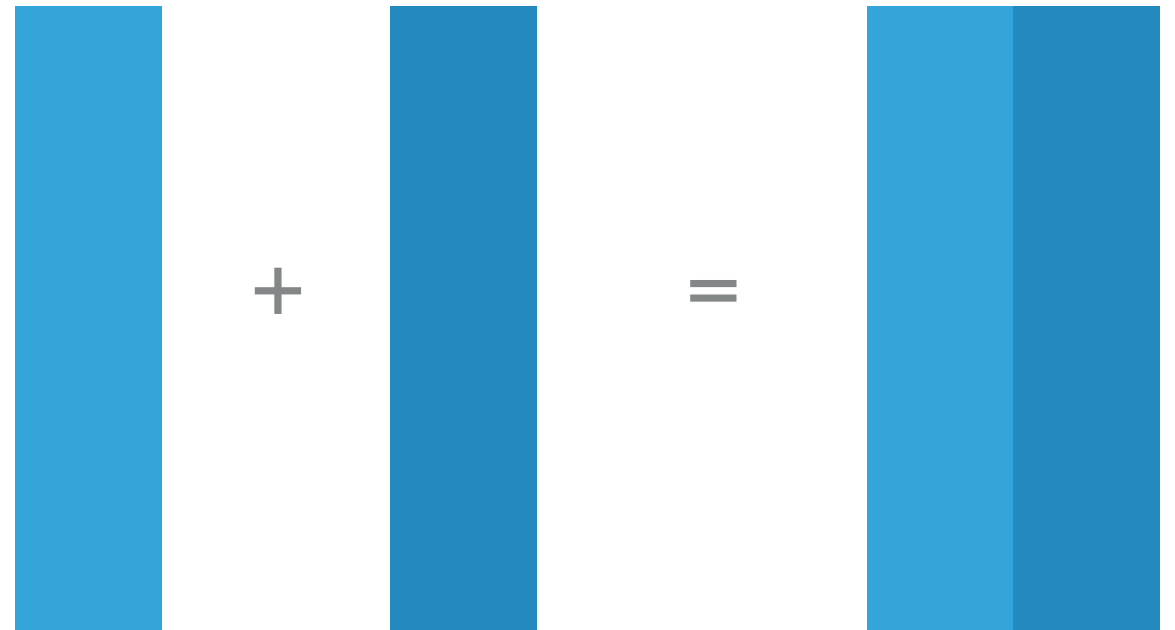
=



DESIGN PRINCIPLE: LAYERING AND SEPARATION

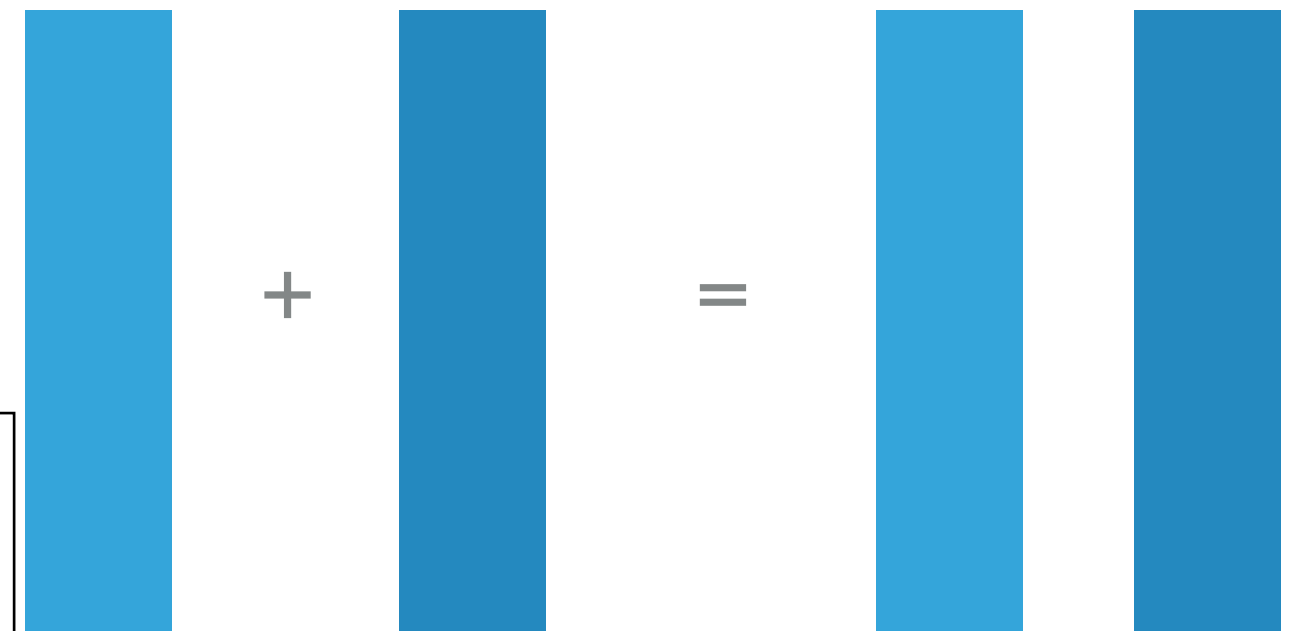
- ▶ Separate and layer information

1+1=2



1+1=3

potential for
negative space to
create clutter

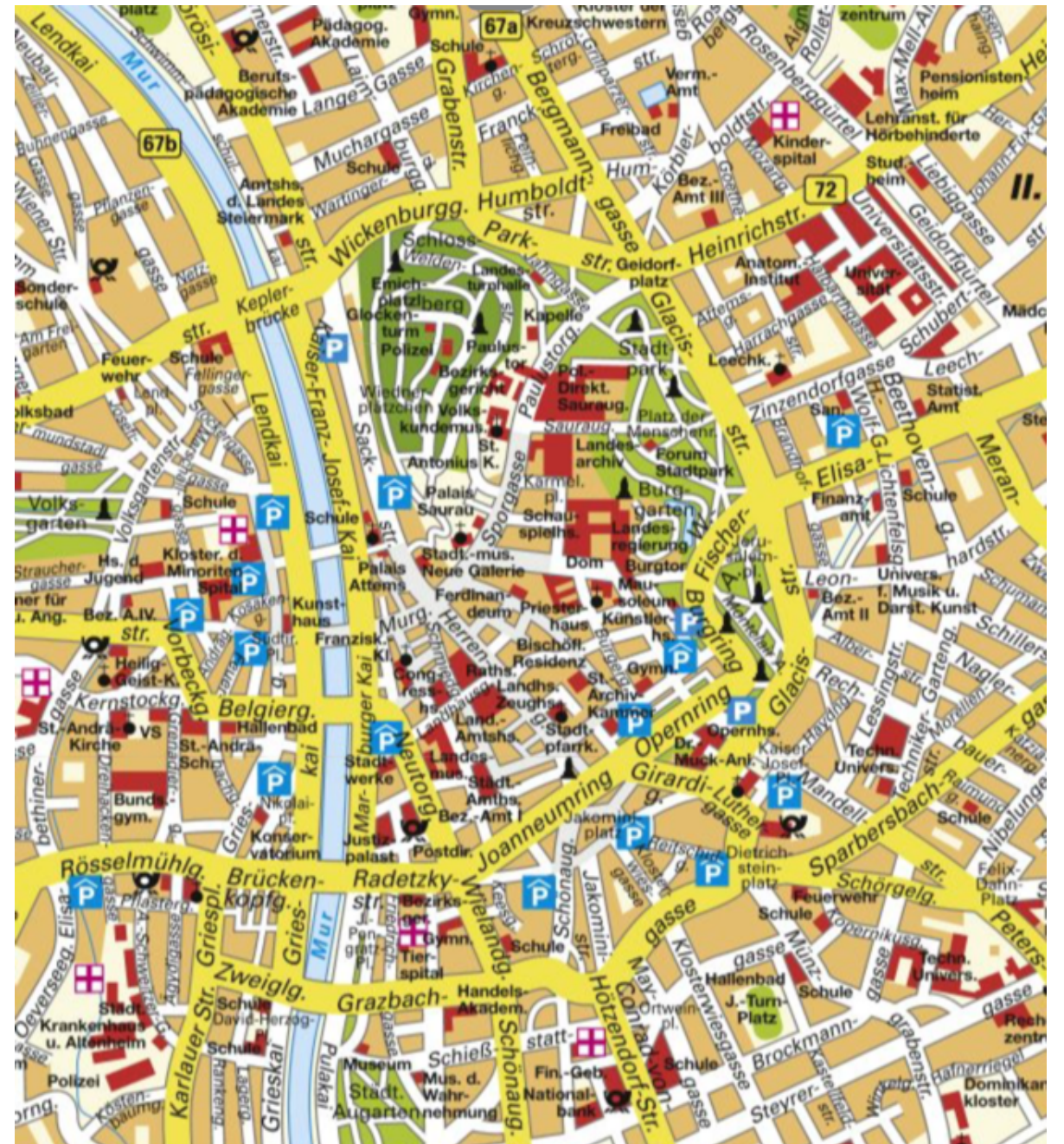


DESIGN PRINCIPLE: LAYERING AND SEPARATION

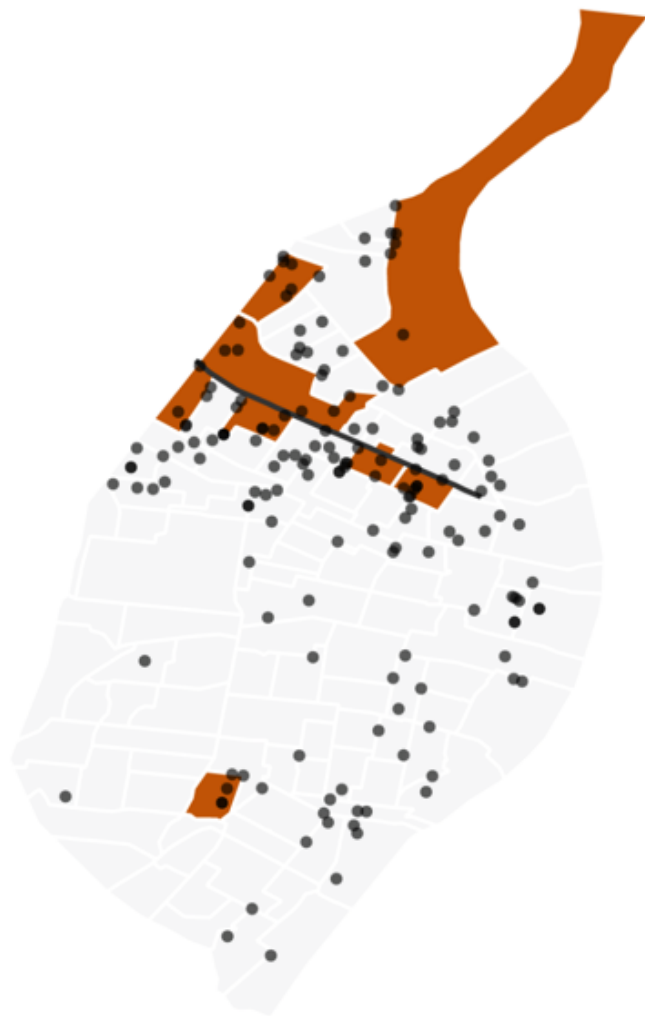
Train No.	3701	3301	3801	3542	3765
	A.M.	A.M.	A.M.	A.M.	A.M.
New York	12:10	12:40	1:30	3:52	4:50
Newark N.J.	12:24	12:55	1:44	4:07	5:04
North Elizabeth					
Elizabeth	12:33	1:03	1:51		5:11
Linden	12:36		1:56		5:16
North Railway					
Rahway	12:40	1:11	2:00		5:20

	am				
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DESIGN PRINCIPLE: LAYERING AND SEPARATION



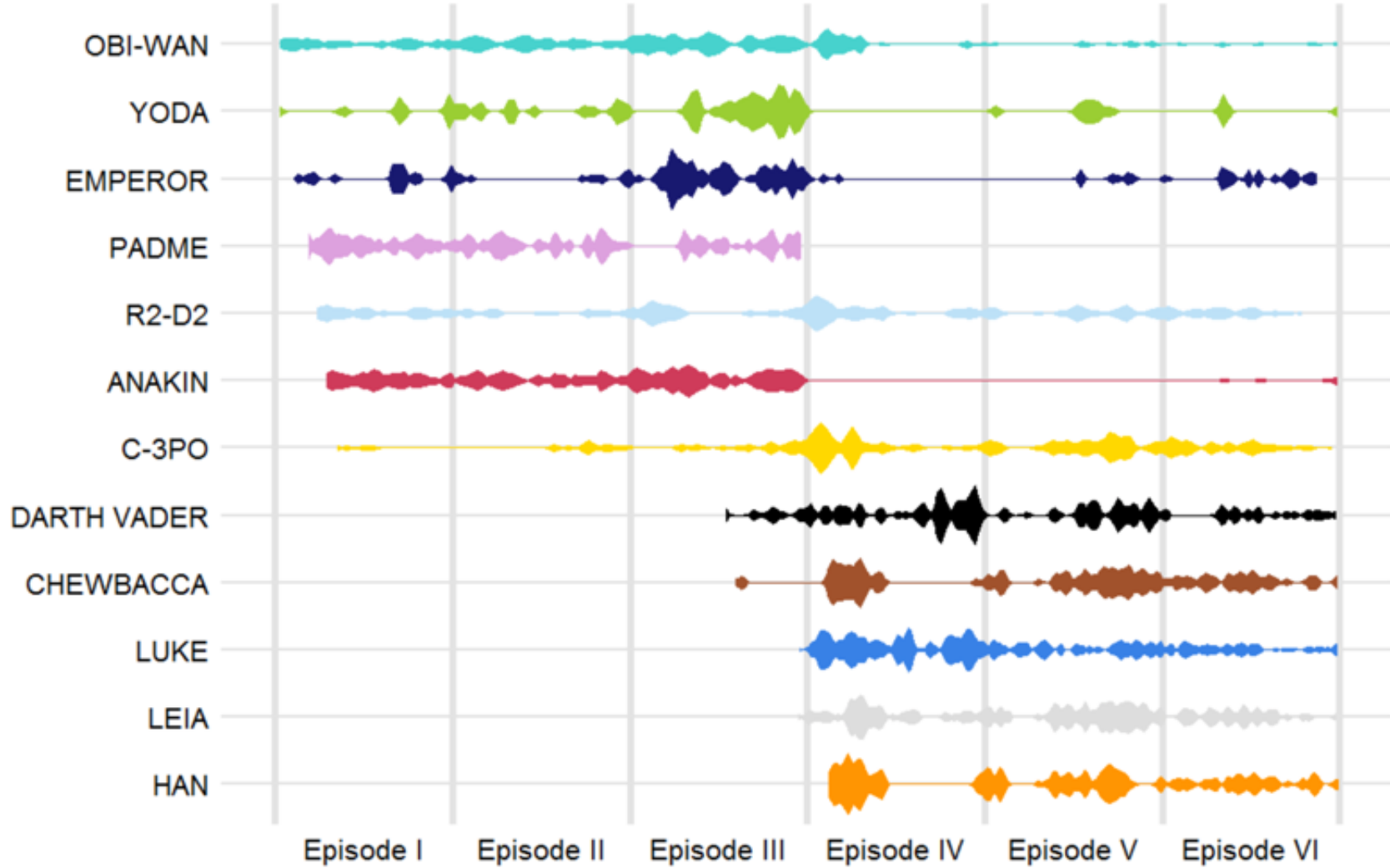
DESIGN PRINCIPLE: LAYERING AND SEPARATION



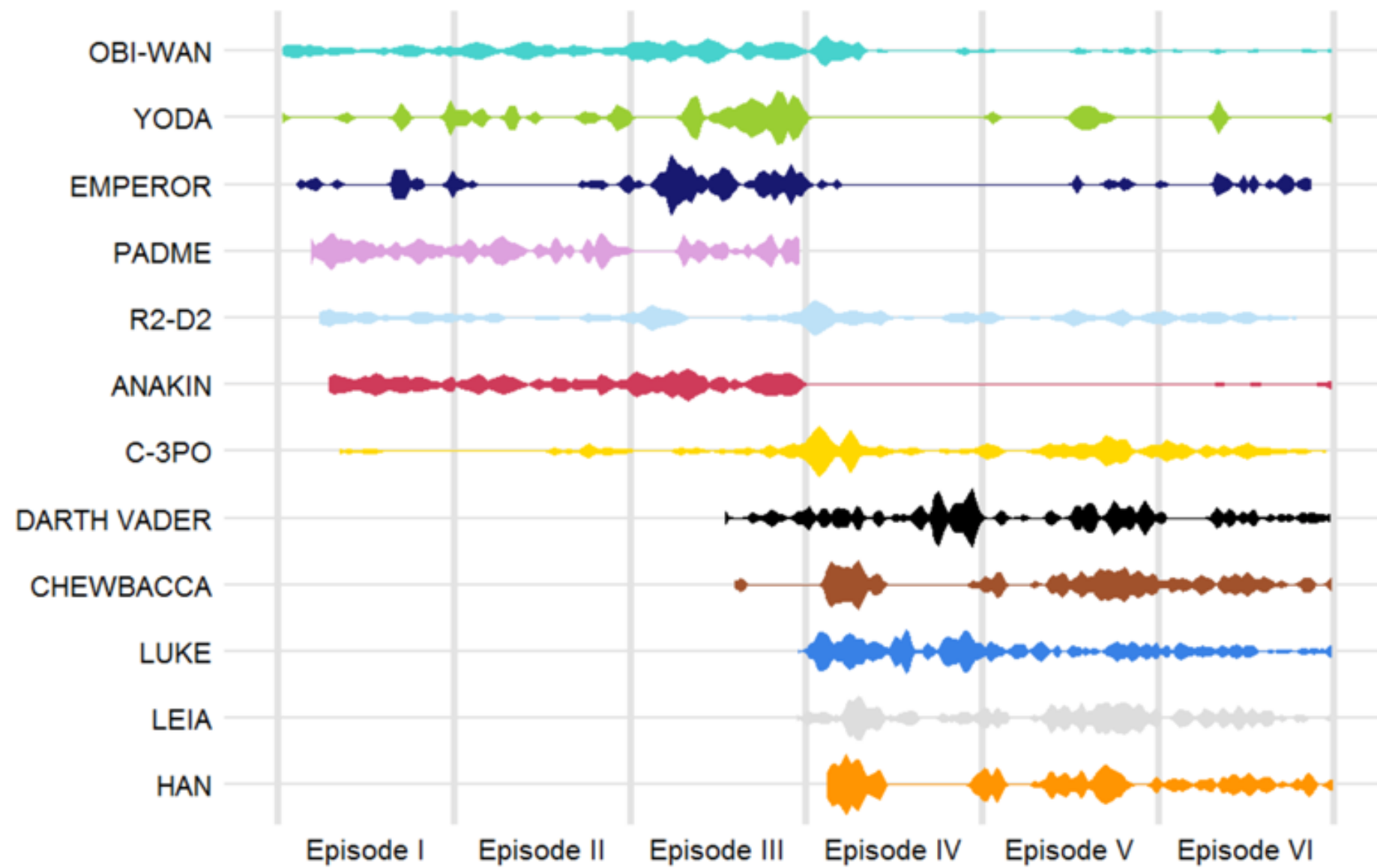
St. Louis

- Census tracts with more than five gun homicides in 2015
- 2015 gun homicide incident
- Natural Bridge Rd.

The Guardian



E. Gabasova





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used in this presentation:

mostly from Microsoft Clipart circa 2009
(unless otherwise specified)
Note references are linked to images in PDF
version of this presentation.